

LABORATORY WORKBOOK
For
BASIC ELECTRICAL ENGINEERING
I Year I Semester
Department of
COMPUTER SCIENCE ENGINEERING [CSE(AIML)]
2025-2026



...Striving Towards Perfection

Geethanjali College of Engineering & Technology
(AUTONOMOUS)

(Accredited by NAAC with 'A+' Grade, Approved by AICTE, Affiliated to JNTUH)

Cheeryal (V), Keesara (M), Medchal Dist. Telangana – 501 301, INDIA

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Geethanjali College of Engineering and Technology (Autonomous)

Accredited by NAAC with A⁺ Grade; B.Tech. CSE, EEE, ECE accredited by NBA

Sy. No: 33 & 34, Cheeryal (V), Keesara (M), Medchal District, Telangana – 501301

LABORATORY WORKBOOK

NAME: _____

ROLL NO: _____

BRANCH: _____ **SECTION:** _____

ACADEMIC YEAR: _____



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Sy. No: 33 & 34, Cheeryal (V), Keesara (M), Medchal District, Telangana – 501301

Certificate

This is to certify that **Mr./Ms.** _____, bearing
Roll No. _____, has successfully completed the prescribed
experiments in the _____ **Laboratory**,
as per the curriculum and guidelines of the College Board of Studies. The student has
completed _____ **out of** _____ **experiments** during the Academic Year _____.

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COURSE COORDINATOR

LECTURER IN-CHARGE

INTERNAL EXAMINER

EXTERNAL EXAMINER

VISION OF THE INSTITUTE

Geethanjali visualizes dissemination of knowledge and skills to students, who would eventually contribute to wellbeing of the people of the nation and global community.

MISSION OF THE INSTITUTE

1. To impart adequate fundamental knowledge in all basic sciences and engineering, technical and Interpersonal skills to students.
2. To bring out creativity in students that would promote innovation, research and entrepreneurship.
3. To Preserve and promote cultural heritage, humanistic and spiritual values promoting peace and harmony in society.

VISION OF THE DEPARTMENT

To produce globally competent and socially responsible computer science engineers contributing to the advancement of engineering and technology which involves creativity and innovation by providing excellent learning environment with world class facilities.

MISSION OF THE DEPARTMENT

1. To be a centre of excellence in instruction, innovation in research and scholarship, and service to the stakeholders, the profession, and the public.
2. To prepare graduates to enter a rapidly changing field as a competent computer science engineer.
3. To prepare graduate capable in all phases of software development, possess a firm understanding of hardware technologies, have the strong mathematical background necessary for scientific computing, and be sufficiently well versed in general theory to allow growth within the discipline as it advances.
4. To prepare graduates to assume leadership roles by possessing good communication skills, the ability to work effectively as team members, and an appreciation for their social and ethical responsibility in a global setting.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

Program Educational Objectives (PEOs) are broad statements that describe what graduates are expected to attain within a few years of graduation. The PEOs for Computer Science and Engineering (Artificial Intelligence and Machine Learning) graduates are:

PEO-I. To provide graduates with a good foundation in mathematics, sciences and engineering fundamentals required to solve engineering problems that will facilitate them to find employment in industry and / or to pursue postgraduate studies with an appreciation for lifelong learning.

PEO-II. To provide graduates with analytical and problem-solving skills to design algorithms, other hardware / software systems, and inculcate professional ethics, inter-personal skills to work in a multi-cultural team.

PEO-III. To facilitate graduates, get familiarized with state-of-the-art software / hardware tools, imbuing creativity and Innovation that would enable them to develop cutting-edge technologies of multidisciplinary nature for societal development.

KNOWLEDGE AND ATTITUDE PROFILE (WK):

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.

WK2: Conceptually based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK7: Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.

WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK9: Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

PROGRAM OUTCOMES (POs):

1. Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
2. Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
3. Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5) .
4. Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
5. Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
6. The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
7. Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
8. Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
9. Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
10. Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
11. Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Demonstrate competency in Programming and problem-solving skills and apply those skills in solving computing problems

PSO2: Select appropriate programming languages, Data structures and algorithms in combination with modern technologies and apply them in developing innovative solutions

PSO3: Demonstrate adequate knowledge in the concepts and techniques of artificial intelligence and machine learning, apply them in developing intelligent systems to solve real world problems

Course overview:

The Basic Electrical Engineering lab course has been designed and developed to ensure that the fundamentals of this course are well understood by students of all Engineering branches. Simultaneously, fundamentals of important topics have been discussed to provide a foundation for the study of advanced topics, by students of various engineering disciplines in their subsequent programmes of study.

Students will get practical exposure of theoretical concepts viz., KCL, KVL, Thevenin's, Nortons and Super Position theorems and will understand the function of transformers and Basic DC Machines and their working in detail during laboratory sessions.

This course ideally meets the requirements of the first level course in 'Basic Electrical Engineering Laboratory'. It is firmly believed that this course will help students to overcome their initial apprehensions and initiate a life-long affair with electrical and electronics engineering.

It also presents a clear and concise exposition of the principles and applications of electrical and electronics engineering

Course Outcomes: On completion of the course, the student would be able to

CO1. Get an exposure to basic electrical laws and theorems

CO2. Verify the basic Electrical circuits through different experiments.

CO3. Evaluate the performance calculations of Electrical Machines and Transformers through various testing methods.

CO4. Analyze the transient responses of R, L and C circuits for different input conditions.

CO -PO mapping:

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	-	-	-	-	1	2	-	1
CO2	3	3	1	-	-	-	-	1	1	-	1
CO3	3	3	1	1	-	-	-	2	1	-	-
CO4	3	3	-	-	1	-	-	1	1	-	-

Indication of Levels: 3-High, 2-Medium, 1-Low

CO's and PSO's mapping.

S.NO	COURSE OUTCOMES	PSO1	PSO2	PSO3
1	Get an exposure to basic electrical laws and theorems	1	1	1
2	Verify the basic Electrical circuits through different experiments.	1	1	1
3	Evaluate the performance calculations of Electrical Machines and Transformers through various testing methods.	1	1	1
4	Analyse the transient responses of R, L and C circuits for different input conditions.	1	1	1

Indication of Levels: 3-High, 2-Medium, 1-Low

Guidelines to Write Workbook

- Components related data to be entered.
- Circuit diagrams, Model graphs, Observations table, Calculations table to be completed.
- Theoretical and model calculations to be written in the provided place.
- Discussion questions to be answered.

Instructions to Laboratory Teachers:

1. Lab records submitted for the lab work are to be checked and signed before the next lab session.
2. Students should be instructed to switch ON the power supply after the connections are checked by the lab assistant /teacher.
3. The promptness of submission should be strictly insisted by awarding the marks accordingly.
4. Ask discussion questions at the end of the experiment.
5. Do not allow students who come late to the lab class.
6. Encourage the students to do the experiments innovatively.
7. Fill continuous Evaluation sheet, on continuous basis.
8. Ensure that the students are dressed in a formal way.

Instructions to the Students:

2. Students should report to the labs concerned as per the timetable schedule.
3. Students who turn up late to the labs will in no case be permitted to perform the experiment scheduled for the day.
4. After completion of the experiment, certification of the staff concerned in-charge in the workbook is necessary.
5. The record of observations along with the detailed experimental procedure of the experiment performed in the immediate last session should be submitted and certified by the staff member in-charge.
6. Not more than three students in a group are permitted to perform the experiment on a setup.
7. The group-wise division made in the beginning should be adhered to, and no mix up of students among different groups will be permitted later.
8. The components required pertaining to the experiment should be collected from stores in charge after duly filling in the requisition form.
9. When the experiment is completed, students should disconnect the setup made by them and should return all the components/instruments taken for the purpose.
10. Any damage of the equipment or burn-out of components will be viewed seriously either by putting penalty or by dismissing the total group of students from the lab for the semester/year.
11. Students should be present in the labs for the total scheduled duration.
12. Students are required to prepare thoroughly to perform the experiment before coming to Laboratory.
13. Procedure sheets/data sheets provided to the students' groups should be maintained neatly and to be returned after the experiment.

25EE11L01-BASIC ELECTRICAL ENGINEERING LAB

B. Tech. I Year I Sem

COMPUTER SCIENCE ENGINEERING [CSE(AIML)]

L	T	P/D	C
-	-	2/-	1

Prerequisite(s): None.

Course Outcomes: On completion of the course, the student would be able to

- CO1. Get an exposure to basic electrical laws and theorems
- CO2. Verify the basic Electrical circuits through different experiments.
- CO3. Evaluate the performance calculations of Electrical Machines and Transformers through various testing methods.
- CO4. Analyse the transient responses of R, L and C circuits for different input conditions.

LIST OF EXPERIMENTS:

1. Verification of KVL and KCL
2. Verification of Thevenin's and Norton's theorem
3. Verification of Superposition theorem
4. Transient response of series RL and RC circuits for DC excitation
5. Resonance in Series RLC circuit
6. Calculations and verification of Impedance and Current of RLC series circuits
7. Measurement of Voltage, Current and Real Power in primary and Secondary Circuits of a Single-Phase Transformer
8. Load Test on Single Phase Transformer (Calculate Efficiency and Regulation)
9. Performance Characteristics of a DC Shunt Motor
10. Torque-Speed Characteristics of a Three-phase Induction Motor.

ADDITIONAL EXPERIMENTS:

1. Three Phase Transformer: Verification of Relationship between Voltages and Currents (Star-Delta, Delta-Delta, Delta-star, Star-Star)
2. Measurement of Active and Reactive Power in a balanced Three-phase circuit

EVALUATION PATTERN

For laboratory course(s), there shall be a Continuous Internal Evaluation (CIE) during the semester for 40 marks, and Semester End Examination (SEE) for 60 marks.

A detailed break up of 40 marks for CIE is given below:

- 1) A write-up on day-to-day experiment in the laboratory (in terms of aim, components/procedure, expected outcome) which shall be evaluated for 10 marks.
- 2) 10 marks for viva-voce (or) tutorial (or) case study (or) application (or) poster presentation of the course concerned.
- 3) Internal practical examination conducted by the laboratory teacher concerned shall be evaluated for 10 marks.
- 4) The remaining 10 marks are for Laboratory Project, which consists of the Design (or) Software / Hardware Model Presentation (or) App Development (or) Prototype Presentation submission which shall be evaluated after completion of laboratory course and before semester end practical examination.

External Examination Assessment: 60 Marks

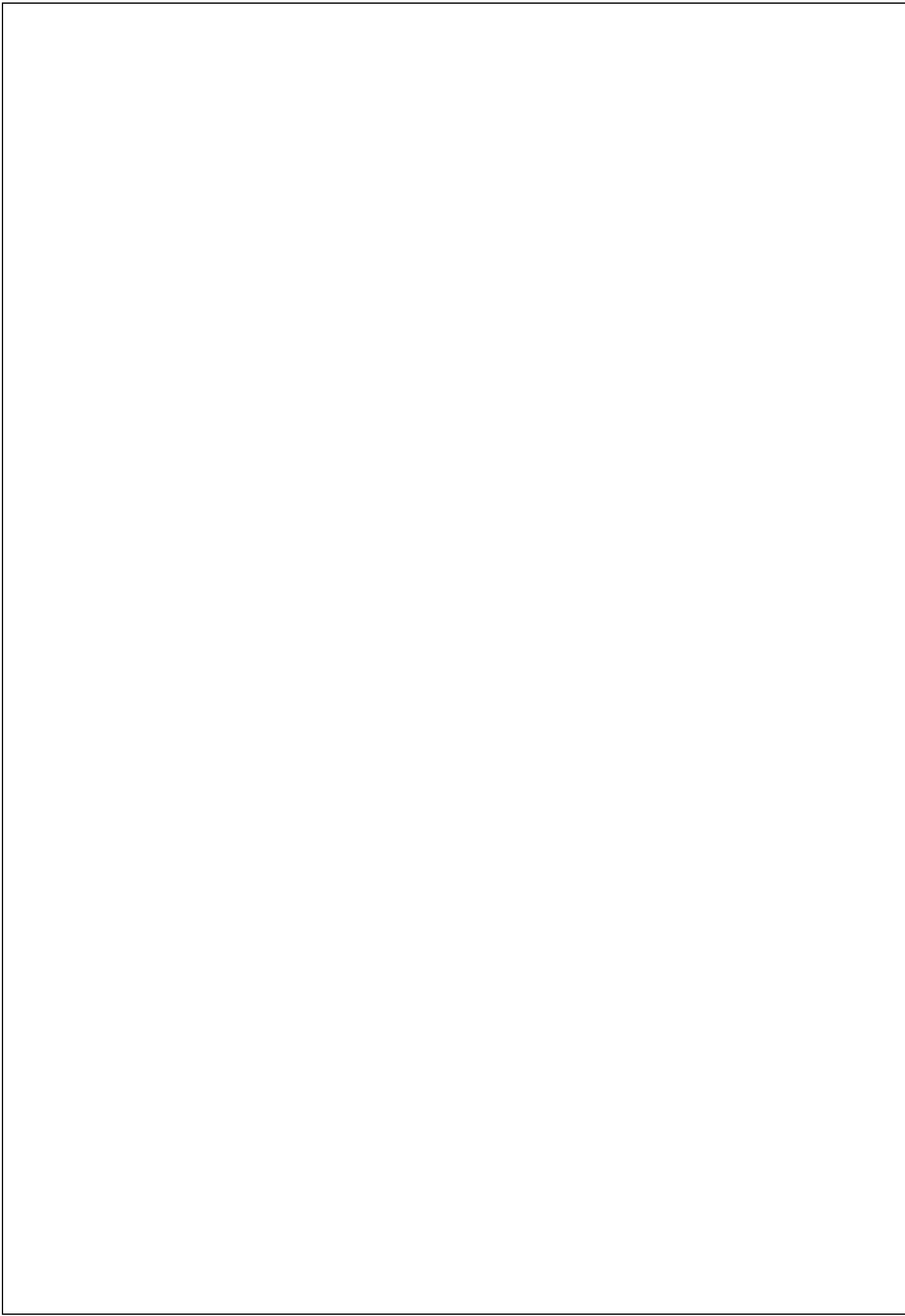
The Semester End Examination shall be conducted with an external examiner and the laboratory teacher. The external examiner shall be appointed from the other reputed colleges which will be decided/approved by the examination branch/Chief Controller of Examinations of the Institution.

In the Semester End Examination held for 3 hours, total 60 marks are divided and allocated as shown below:

1. 10 marks for write-up
2. 15 for experiment/program
3. 15 for evaluation of results
4. 10 marks for presentation on another experiment/program in the same laboratory course and
5. 10 marks for viva-voce on concerned laboratory course

Index

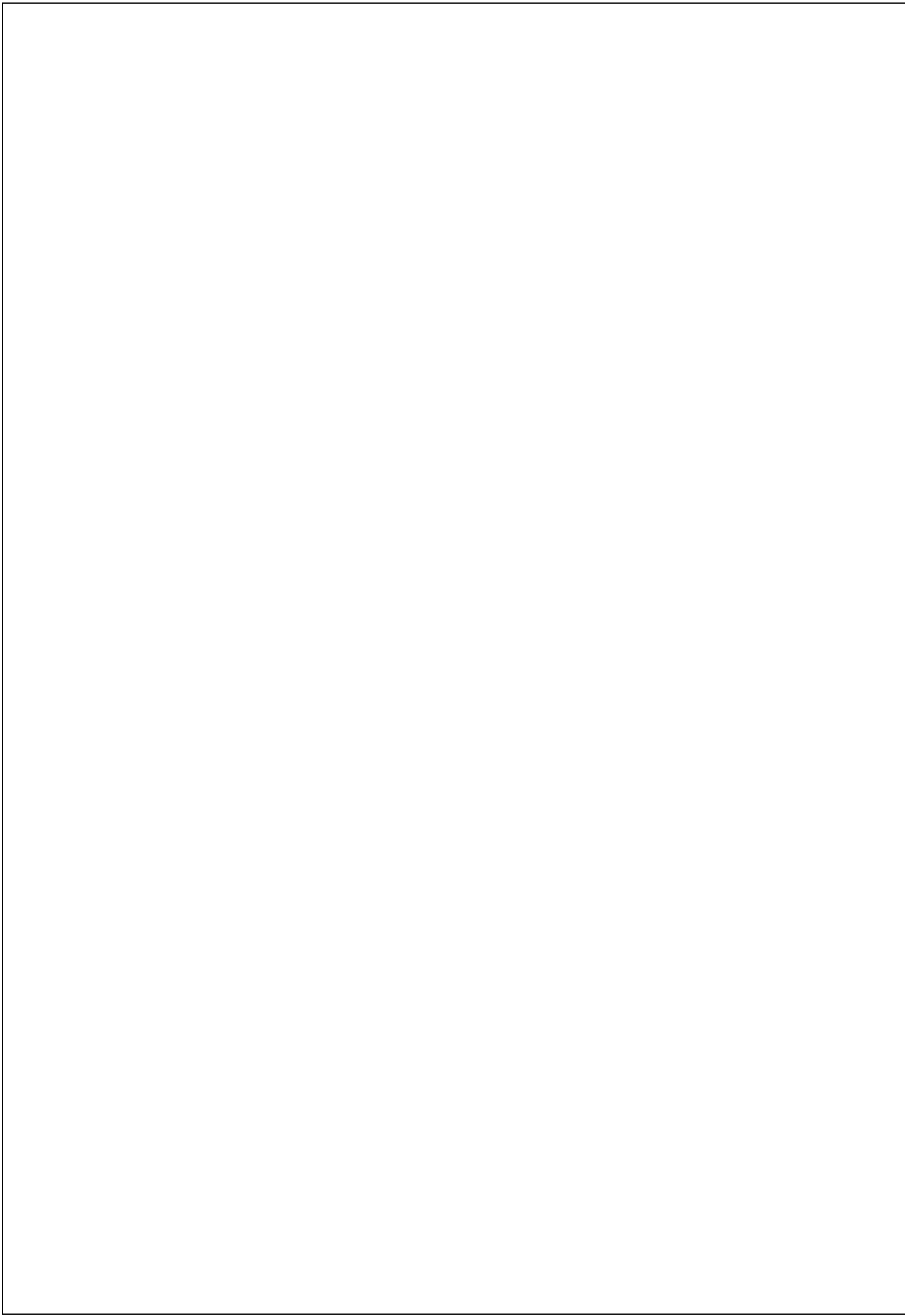
S. No.	Name of the Experiment	Page No	Date of Experiment	Marks Awarded	Signature of Faculty	Remarks (if any)
			Date of Submission			
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						



.....ΩφσδτΠΣ

Cycle- I

.....ΩφσδτΠΣ



Experiment No: _____

Date: _____

VERIFICATION OF KVL AND KCL

Objective: To verify Kirchhoff's Voltage law and Kirchhoff's Current law theoretically and practically.

Apparatus:

S. NO	Name of the Equipment	Range	Type	Quantity
1	Dual Channel Regulated Power Supply	(0-30)V/2A	Digital	1
2	Resistors	1k Ω , 1.5k Ω , 2.2k Ω	Carbon composition or film	1 each
3	Voltmeters	(0-30)V	Digital	3
4	Ammeters	(0-200)mA	Digital	3
5	Bread Board	-	-	1
6	Connecting Wires	-	-	As per requirement

Theory:

Circuit: A DC circuit is a closed loop conducting lane in which an electrical current flow.

Path: A single lane is used to connect the sources or elements.

Node: A node is a connection in a circuit where multiple elements are connected together, and it is denoted with a dot.

Branch: a branch is a single or collection of elements which are connected between two nodes like resistors or a source

Loop: A loop in a circuit is a closed path, where no circuit element or node is met more than once.

Mesh: A mesh doesn't contain any closed path, but it is a single open loop, and it does not contain any components inside a mesh.

Statement:

KCL: The algebraic sum of currents at a node or junction is Zero.

KVL: The algebraic sum of voltages or emfs and the voltage drops in a closed loop is zero.

Circuit Diagram: For KVL

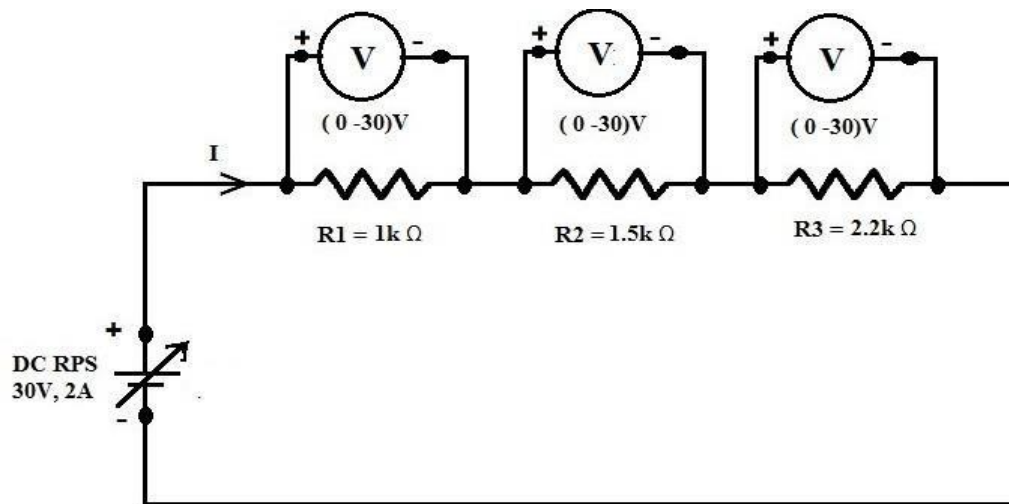


Fig 1.1 Circuit diagram for KVL

Procedure:

1. Give the connections as per the circuit diagram 1.1.
2. Set current limit knob to maximum position and Voltage knobs to minimum position.
3. Switch on the supply and switch on DC RPS supply toggle switch.
4. Set a particular voltage value in RPS.
5. Note down the corresponding voltmeter readings.
6. Repeat the step 4 and 5 for different voltages.
7. After completion of experiment set voltage knobs to minimum position and switch off DC RPS toggle switch and supply switch.

Observation table for Practical values: KVL

S. No	Supply Voltage V_s (in volts)	Voltage Drops (in volts)			Total Voltage (in volts) $V_s = V_1 + V_2 + V_3$
		V_1	V_2	V_3	

Observation table for Theoretical values:KVL

S. No	Supply Voltage V_s (in volts)	Voltage Drops (in volts)			Total Voltage (in volts) $V_s = V_1 + V_2 + V_3$
		V_1	V_2	V_3	

Theoretical Calculations for KVL:

Circuit Diagram: For KCL

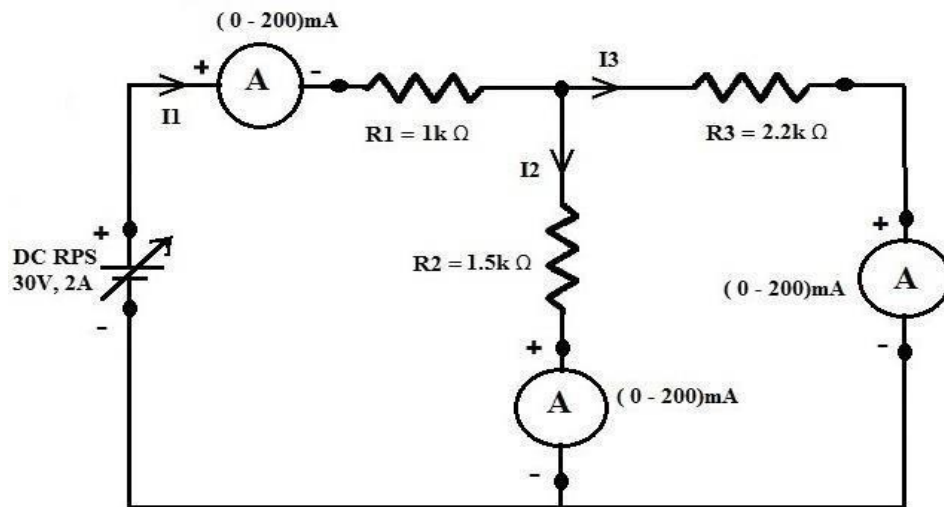


Fig 1.2 Circuit diagram for KCL

Procedure:

1. Give the connections as per the circuit diagram 1.2.
2. Set current limit knob at maximum position and Voltage knobs at minimum position.
3. Switch on the supply and switch on DC RPS supply toggle switch.
4. Set a particular voltage value in RPS.
5. Note down the corresponding Ammeter readings.
6. Repeat the step 4 and 5 for different voltages.
7. After completion of experiment set voltage knobs to minimum position and switch off DC RPS toggle switch and supply switch.

Observation table for Practical values: KCL

S. No	Supply Voltage V_s (in volts)	Branch Currents (in mA)			Total Current $I_1=I_2+I_3$ (in mA)
		I_1	I_2	I_3	

Observation table for Theoretical values: KCL

S. No	Supply Voltage Vs (in volts)	Branch Currents (in mA)			Total Current $I_1=I_2+I_3$ (in mA)
		I_1	I_2	I_3	

Theoretical Calculations for KCL:

Result:**Discussion Questions:**

1. The internal resistance of an ideal voltage source is_____.
2. The internal resistance of an ideal current source is_____.
3. What is a Linear Network?
4. What are the types of sources?
1. What are the types of active and passive elements?
2. What is source transformation?
3. State ohm's law?

Outcome:

By completing this experiment student would be able to know the concept of Kirchhoff's Laws and able to solve the problems using these laws.

EXPERIMENT NO:

DATE:

EXPERIMENTAL VERIFICATION OF THEVENIN'S AND NORTON'S THEOREM

Objective: To verify Thevenin's theorem and Norton's theorem, both theoretically and practically.

Apparatus:

S. No	Name of the equipment	Range	Type	Quantity
1	Ammeter	0-200mA	Digital	2
2	Voltmeter	(0-30) V	Digital	1
3	Bread board			1
4	Resistors	1k Ω , 1.5k Ω , 2.2k Ω , 1k Ω	Carbon film	1 each
5	RPS	0-30V/2A	DC	1
6	Digital Multimeter	-	Digital	1
7	Connecting wires			as per need

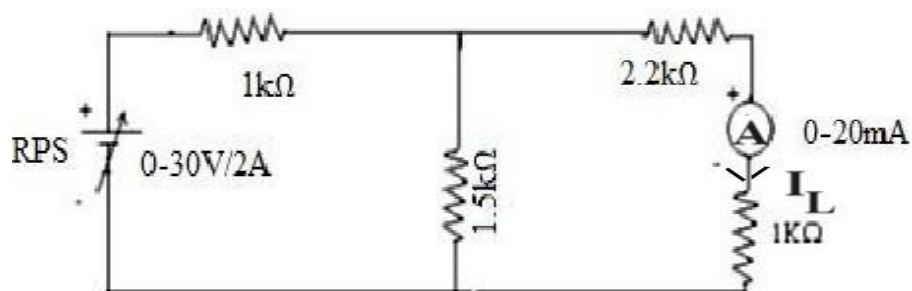
Theory:

Statement of Thevenin's Theorem: Any linear, bilateral electrical network containing voltage and current sources and resistors can be replaced at a pair of terminals by an equivalent voltage source V_{th} in series with an equivalent resistance R_{th} . Here, V_{th} is the open-circuit voltage measured across the load terminals and R_{th} is the equivalent resistance seen between the two terminals with all independent energy sources replaced by their internal resistances (voltage sources shorted, current sources opened).

Statement of Norton's Theorem: Any linear, bilateral electrical network containing voltage and current sources and resistors can be replaced at a pair of terminals by an equivalent current source I_N in parallel with an equivalent resistance R_N . Here, I_N is the short-circuit current flowing between the load terminals, and R_N is the equivalent resistance seen between the two terminals with all independent energy sources replaced by their internal resistances (i.e., voltage sources are short-circuited and current sources are open-circuited).

Experimental Verification of Thevenin's Theorem:

Fig 2.1: To find Load Current (I_L) from the original circuit :



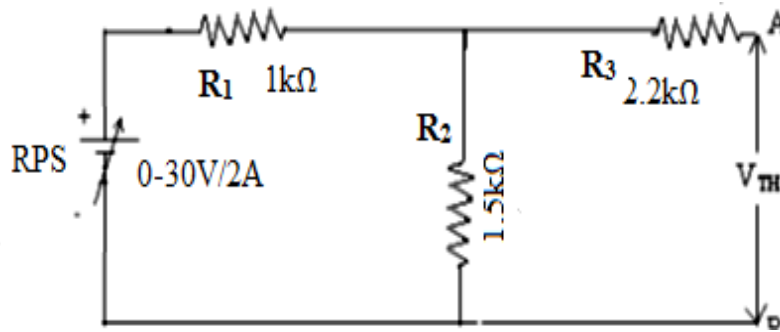
Procedure to Find Load Current (I_L):

1. Connect the circuit as shown in Fig 2.1.
2. Switch ON the DC power supply and set an appropriate voltage (say $V_s = 5V$) using the DC RPS.
3. Note the load current (I_L) from the ammeter.
4. Repeat the procedure for different values of V_s .
5. Record both the practical and theoretical values of I_L in the observation table.

Table 2.1: Tabular column for I_L values:

S. No.	$V_s(V)$	$I_L(mA)$	
		Theoretical	Practical
1			
2			
3			

Fig 2.2: To find Thevenins Voltage (V_{TH}):



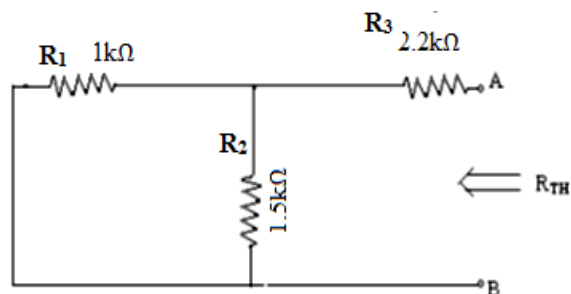
Procedure to Find Thevenin Voltage (V_{TH})

1. Remove the load resistor R_L .
2. Connect the circuit as in Fig 2.2.
3. Keep the same supply voltage (e.g., 5V).
4. Measure the open-circuit voltage across terminals AB using voltmeter – this is V_{TH} .
5. Repeat the procedure for different values of V_s .

Table 2.2: Tabular column for noting V_{TH} :

S. No.	$V_s(V)$	$V_{TH}(V)$	
		Theoretical	Practical
1			
2			
3			

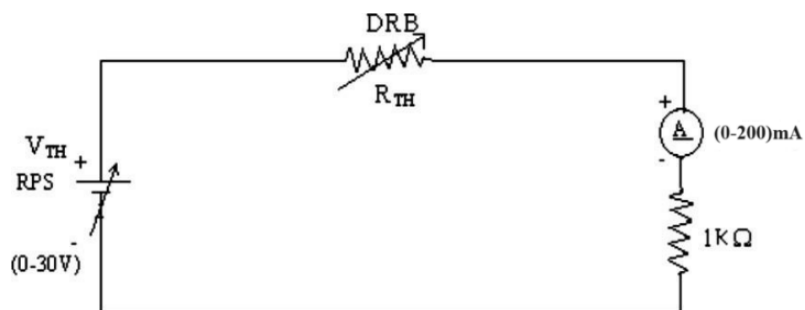
Fig 2.3: To find Thevenin Resistance R_{TH} :



Procedure to Find Thevenin Resistance (R_{TH})

1. Turn OFF the power supply.
2. Replace voltage sources with short circuits (use wires).
3. Measure the resistance across terminals using a multimeter – this is R_{TH} . $R_{TH} = \underline{\hspace{2cm}}$

Fig 2.4: To find load current I_L^1 from Thevenin's equivalent circuit:



Procedure to find I_L^1 :

1. Construct the Thevenin equivalent circuit as shown in Fig 2.4.
2. Apply the V_{TH} voltage using DC RPS.
3. Note the load current (I_L^1) from the ammeter.
4. Compare the value of I_L^1 with the original load current I_L measured in fig:2.1
5. If $I_L^1 \approx I_L$, then Thevenin's Theorem is verified.

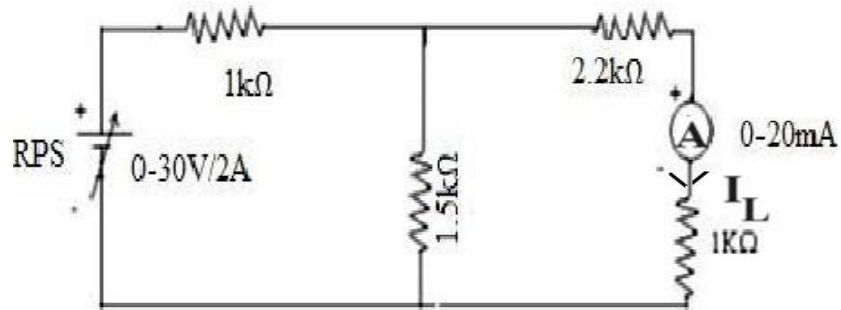
Table 2.4: Tabular column for noting I_L^1

S. No.	$V_{TH}(V)$	$I_L^1(mA)$	
		Theoretical	Practical
1			
2			
3			

Thevenin's Theorem Theoretical Calculations:

Experimental Verification of Norton's Theorem:

Fig 2.1: To find Load Current (I_L) from the original circuit:



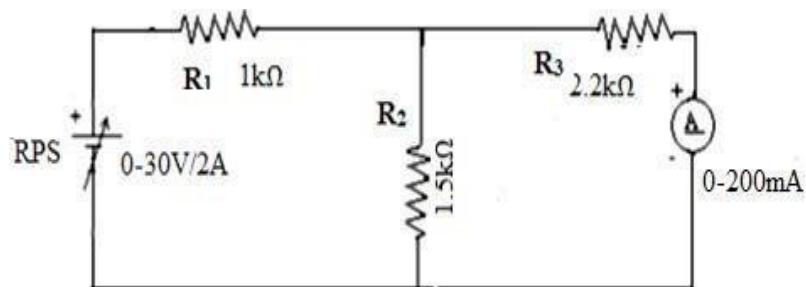
Procedure to Find Load Current (I_L):

1. Connect the circuit as shown in Fig 2.1.
2. Switch ON the DC power supply and set an appropriate voltage (say $V_s = 5V$) using the DC RPS.
3. Note the load current (I_L) from the ammeter.
4. Repeat the procedure for different values of V_s .
5. Record both the practical and theoretical values of I_L in the observation table.

Table 2.1: Tabular column for I_L values:

S. No.	$V_s(V)$	$I_L(mA)$	
		Theoretical	Practical
1			
2			
3			

Fig 2.5: To find Nortons Current (I_N)



Procedure to Find Norton Current (I_N)

1. Remove the load resistor R_L
2. Connect an ammeter across the open terminals (short circuit them).
3. The ammeter reading is the Norton current (I_N).

Table 2.4: Tabular column for noting I_N :

S. No.	Vs(V)	$I_N(\text{mA})$	
		Theoretical	Practical
1			
2			
3			

Nortons Resistance $R_N = R_{TH}$ (Thevenins resistance)

$R_N = \underline{\hspace{2cm}}$

Fig 2.6: To find load current (I_L) from Nortons equivalent circuit:

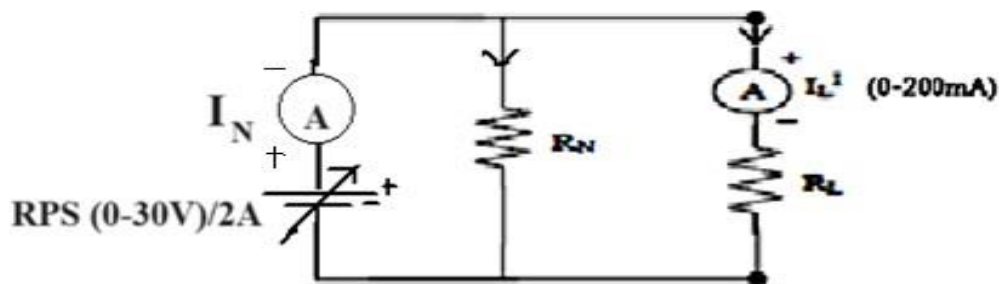


Fig:2.6

Procedure for Finding I_L^1

1. Make the connections as per the circuit shown in the Fig 2.6.
2. Apply the Norton's current to the circuit as shown and note the load current I_L^1 from the ammeter connected to the load terminals.

Table 2.5: Tabular column for noting I_L^1 : for Norton's verification

S. No.	I_N (A)	I_L^1 (mA)	
		Theoretical	Practical
1			
2			
3			

Nortons Theorem Theoretical Calculations:

Result:

Outcome: The student will be able to identify basic circuit components and apply various network theorems, including Thevenin's and Norton's theorems, to analyze and solve practical electrical circuits.

Viva Questions:

1. State Thevenin's theorem.
2. State Norton's theorem.

Experiment No: _____

Date: _____

VERIFICATION OF SUPERPOSITION THEOREM

Objective: To verify Superposition Theorem theoretically and practically.

Apparatus:

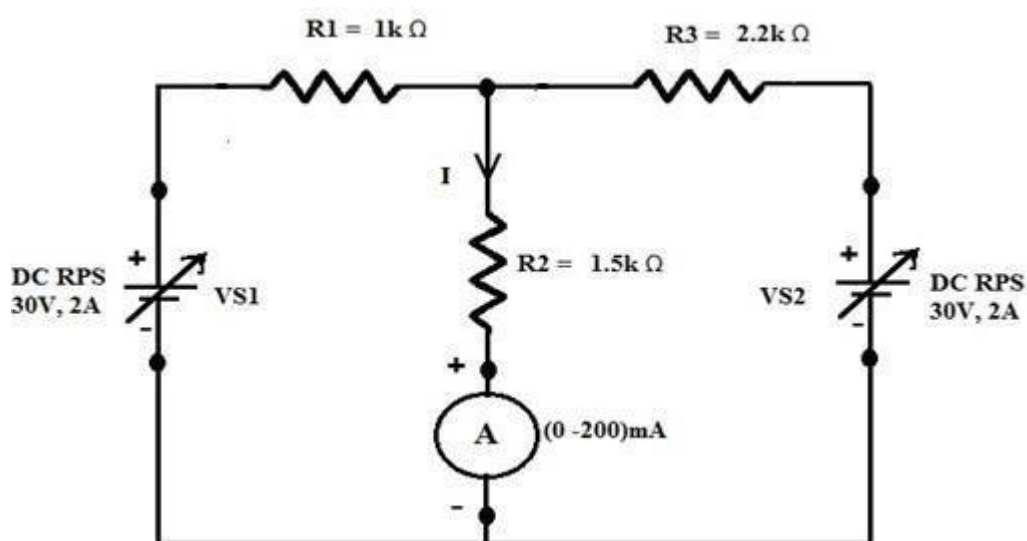
S. NO	Name of the Equipment	Range	Type	Quantity
1	Dual Channel Regulated Power Supply	(0-30)V/2A	Digital	1
2	Resistors	1k Ω , 1.5k Ω , 2.2k Ω	Carbon composition or film	1 each
3	Ammeters	(0-200)mA	Digital	1
4	Bread Board	-	-	1
5	Connecting Wires	-	-	As per requirement

Theory:

Superposition Theorem Statement:

“In any linear bilateral network containing two or more energy sources the response at any element is equal to the algebraic sum of the responses caused by the individual sources.” While considering the effect of individual sources, the other ideal voltage sources and ideal current sources in the network are replaced by short circuit and open circuits respectively, across the terminals. (It means replacing them with their internal resistances or impedances). This theorem is valid only for linear systems. It is also very useful in the analysis of circuits. The superposition theorem can be very useful when the circuit has large number of sources (current or voltage sources), to find the value of current or voltage in any branch of the circuit.

Case 1: Both Sources V_{S1} , V_{S2} are active.



Circuit 2.1: When both the sources are active.

Procedure:

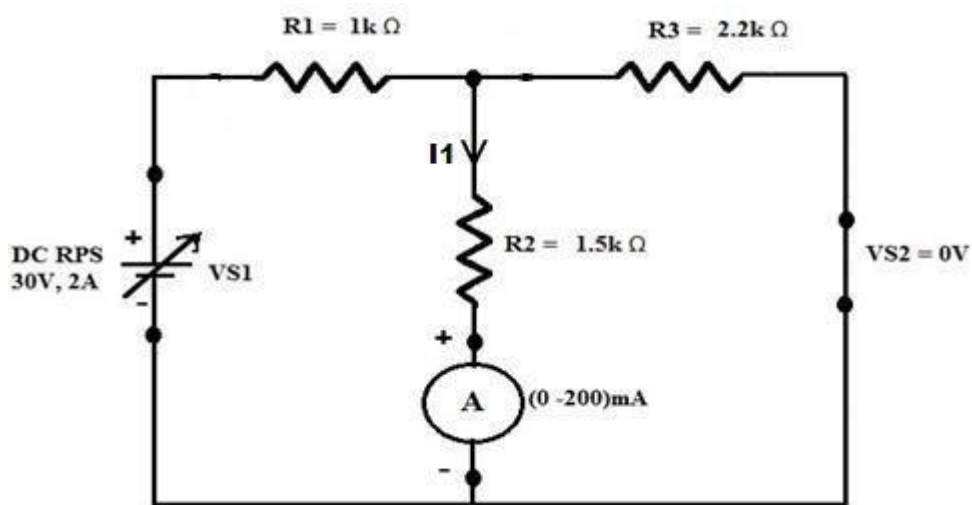
1. Give the connections as the circuit diagram 2.1.
2. Set current limit knob at maximum position and Voltage knobs at minimum position.
3. Switch on the supply and switch on DC RPS supply toggle switch.
4. Set a particular voltage value V_{S1} by using channel-1 and V_{S2} by using channel -2 respectively in RPS.
5. Note down Ammeter reading.
6. Repeat step 4 and 5 for different voltages and note down the ammeter reading.
7. After completion of experiment set voltage knobs to minimum position and switch off DC RPS toggle switch and supply switch.

Observation Table:

S. No	Source voltage V_{S1} (in volts)	Source voltage V_{S2} (in volts)	Practically	Theoretically
			Load current I_L (in mA)	

Theoretical Calculations:

Case 2: Only V_{S1} source is active and V_{S2} is inactive (replaced by short circuit)



Circuit-2.2 : When V_{S1} source is acting alone and V_{S2} is replaced by short circuit

Procedure:

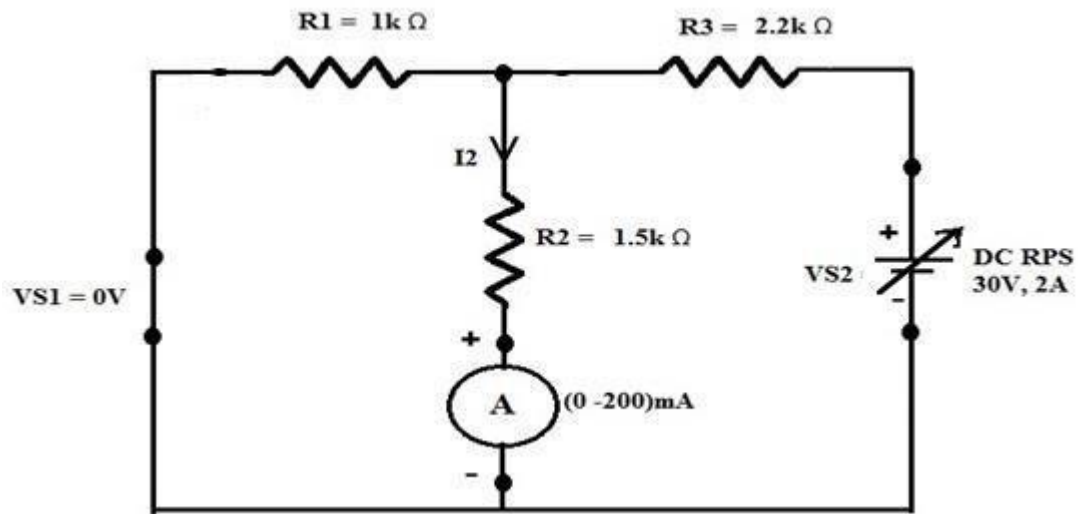
1. Give the connections as the circuit diagram 2.1.
2. Set current limit knob at maximum position and Voltage knobs at minimum position.
3. Switch on the supply and switch on DC RPS supply toggle switch.
4. Set a particular voltage value V_{S1} by using RPS channel -1 and short circuiting V_{S2} .
5. Note down the ammeter reading.
6. Repeat step 4 and 5 for different voltages and note down the ammeter reading.
7. After completion of experiment set voltage knobs to minimum position and switch off DC RPS toggle switch and supply switch.

Observation Table:

S. No	Source voltage V_{S1} (in volts)	Source voltage V_{S2} (in volts)	Practically	Theoretically
			Load current I_L (in mA)	
		0V		
		0V		
		0V		

Theoretical Calculations:

Case 3: Only V_{S2} source is active and V_{S1} is inactive (replaced by short circuit)



Circuit-2.3 : When V_{S2} source is acting alone and V_{S1} is replaced by short circuit

Procedure:

1. Give the connections as the circuit diagram 2.1.
2. Set current limit knob at maximum position and Voltage knobs at minimum position.
3. Switch on the supply and switch on DC RPS supply toggle switch.
4. Set a particular voltage value V_{S2} by using RPS channel -2 and short circuiting V_{S1} .
5. Note down the ammeter reading.
6. Repeat step 4 and 5 for different voltages and note down the ammeter reading.
7. After completion of experiment set voltage knobs to minimum position and switch off DC RPS toggle switch and supply switch.

Observation Table:

S. No	Source voltage V_{S1} (in volts)	Source voltage V_{S2} (in volts)	Practically	Theoretically
			Load current I_L (in mA)	
	0 V			
	0 V			
	0 V			

Theoretical Calculations:

Result:

Discussion Questions

1. What is a short circuit and open circuit?
2. When superposition theorem is used?
3. What is meant by a linear bilateral circuit?
4. Superposition theorem is not applicable in-----
(A) Voltage responses (B) Power responses
(C) Current responses (D) All of the above
5. Superposition theorem is valid only for-----
(A) Linear circuits (B) Non-linear circuits
(C) Both linear and non-linear circuits (D) none
6. What is DC RPS?

Outcome:

By completing this experiment student would be able to know the apply superposition theorem and find out the response (voltage or current).

Experiment No: _____

Date: _____

TRANSIENT RESPONSE OF SERIES 'RL' AND 'RC' CIRCUITS USING DC EXCITATION

Objective:

To study the Transient responses of series RL and RC circuits by applying a DC excitation (Step input voltage).

Apparatus:

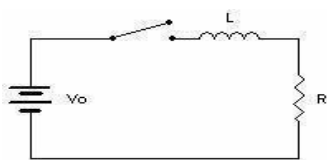
S. No	Name of the Equipment	Range	Type	Quantity
1	Function Generator	(0 – 1MHz)	Digital	1
2	Resistors	1kΩ, 10kΩ	Carbon composition or film	1 each
3	Decade Inductance box (DLB)	1H	-	1
4	Capacitor	0.1μf	Film capacitors	1
5	Cathode ray Oscilloscope (CRO) & CRO Probes	(0 – 30MHz)	Analog	1
6	Bread Board	-	-	1
7	Connecting Wires	-	-	As per requirement

Theory:

Transients in electric circuits occur due to the presence of energy storage elements (i.e., inductors and capacitors).

Series RL Circuit:

If the switch in this circuit is initially open, and then closed at time $t=0+$, the current in this circuit will be described as:



$$i(t) = I_o(1 - e^{\frac{-t}{\tau}})$$

$$v(t) = V_o(e^{\frac{-t}{\tau}})$$

Where: I_o , V_o = initial current and voltage after switch is closed ($t=0^+$) respectively

$\tau = \frac{L}{R}$ is the time constant for the series RL circuit

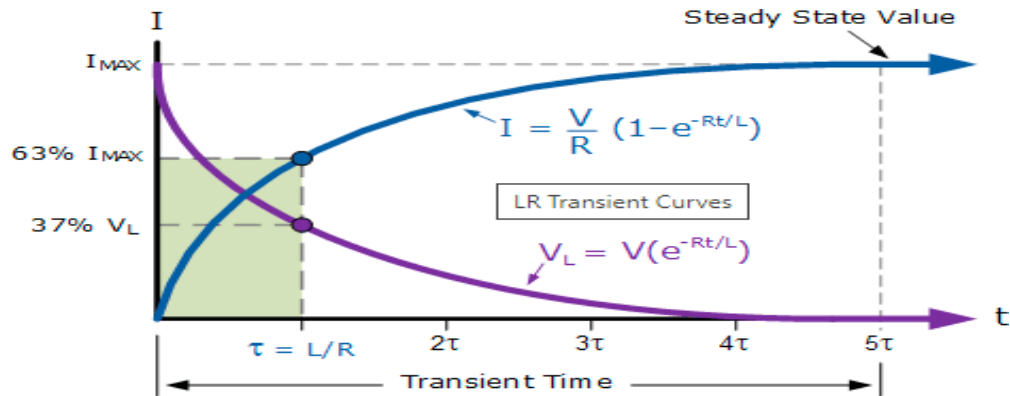
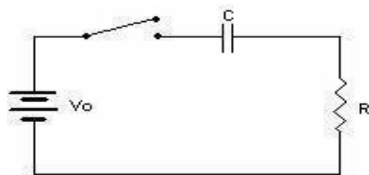


Fig 3.1 Transient response across inductor

Series RC Circuit:

If the switch in this circuit was initially open, and then closed at time $t=0+$, the current in this circuit



$$i(t) = I_o(e^{-\frac{t}{\tau}})$$

$$v(t) = V_o(1 - e^{-\frac{t}{\tau}})$$

Where:

I_o, V_o = initial current and voltage after switch is closed ($t=0^+$) respectively

$\tau = RC$ is the time constant (sec) for the RC series circuit

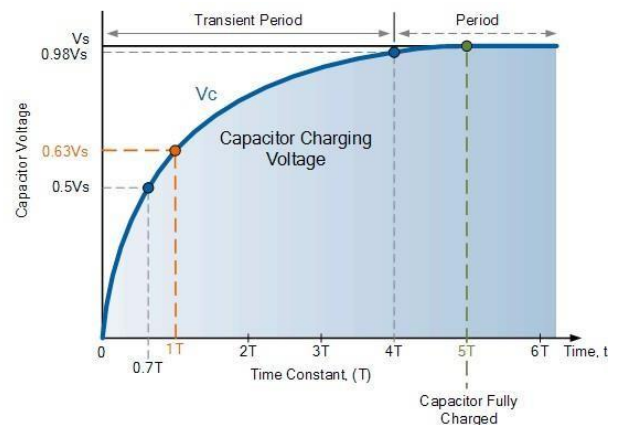
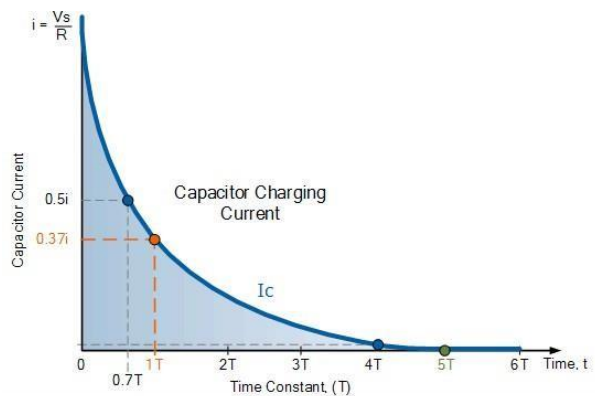


Fig 3.2 Transient response across capacitor

Generally, when the elapsed time exceeds five time constants ($5t$) after switching has occurred, the currents and voltages have reached their final value, which is also called steady-state response.

Procedure:**Series RL Circuit:**

1. Connections are made as per the circuit –1.
2. Turn on Function generator and CRO.
3. Adjust CRO tracer on X-axis using ground(GD) button.Keep the switch in DC mode
4. Set the function generator to deliver a square wave:
 - i. Select function button as square wave, select frequency range as 100Hz and keep frequency knob at maximum position.
 - ii. Keep the Amplitude knob in maximum position.
 - iii. Turn ON and keep DC offset knob in maximum position and adjust the input waveform to X-axis.
5. Set the VOLTS/DIV to 5V and TIME/DIV to 2ms and draw the input waveform in graph.
6. Connections are made as per the circuit –2, Connect CRO probes across Inductor, adjust the output waveform to X-axis and draw in graph.
7. Find the voltage across inductor (V_L) at different time constants (τ) and verify it with theoretical calculations.

Observation from CRO:

When input and voltage waveform across Inductor is observed on CRO. To calculate the voltage and time across Y –axis and X-axis following calculations are done.

To measure time on X-axis:-

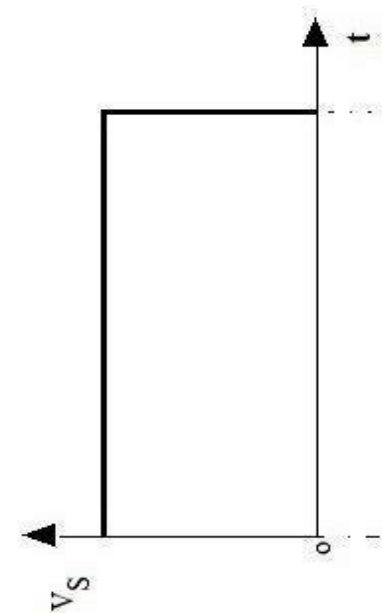
Time(T) : - _____ number of **Horizontal divisions**X _____ $\frac{\text{Time}}{\text{Division}} =$

To measure Voltage on Y-axis:-

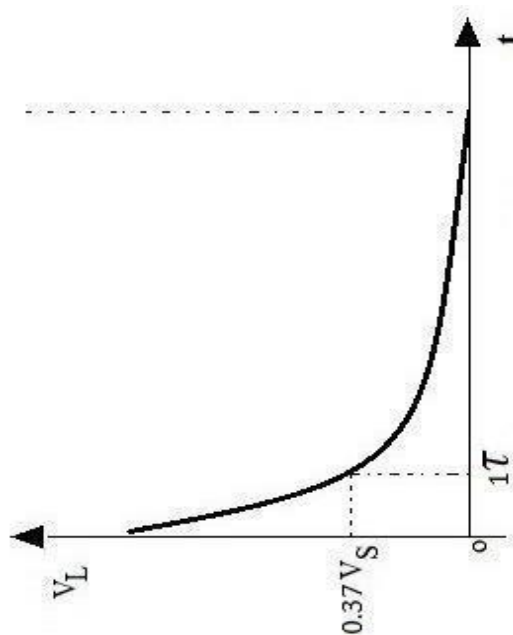
Input Voltage (V_s): - _____ number of **Vertical divisions**X _____ $\frac{\text{Voltage}}{\text{Division}} =$

Series RL Circuit:**Theoretical Calculations:** $V_S = \underline{\hspace{2cm}}$ VVoltage across Inductor: $V_L = V_S (e^{\frac{-t}{\tau}})$ Time constant $\therefore \tau = \frac{L}{R} =$ Calculate V_L (i) $t = 1 \tau$; $V_L =$ (ii) $t = 2 \tau$; $V_L =$ (iii) $t = 3 \tau$; $V_L =$ (iv) $t = 4 \tau$; $V_L =$ (v) $t = 5 \tau$; $V_L =$

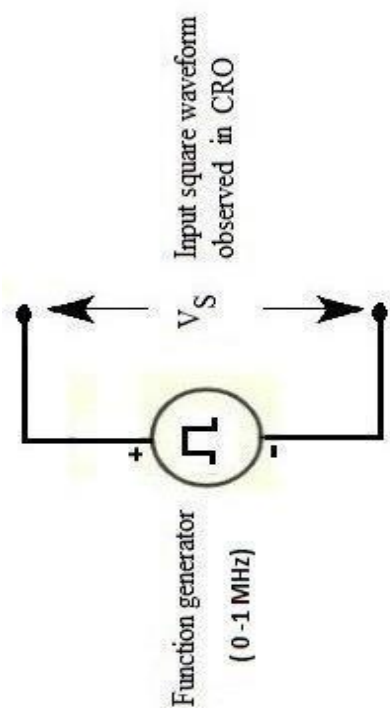
Circuit Diagram 3.1: Series RL Circuit



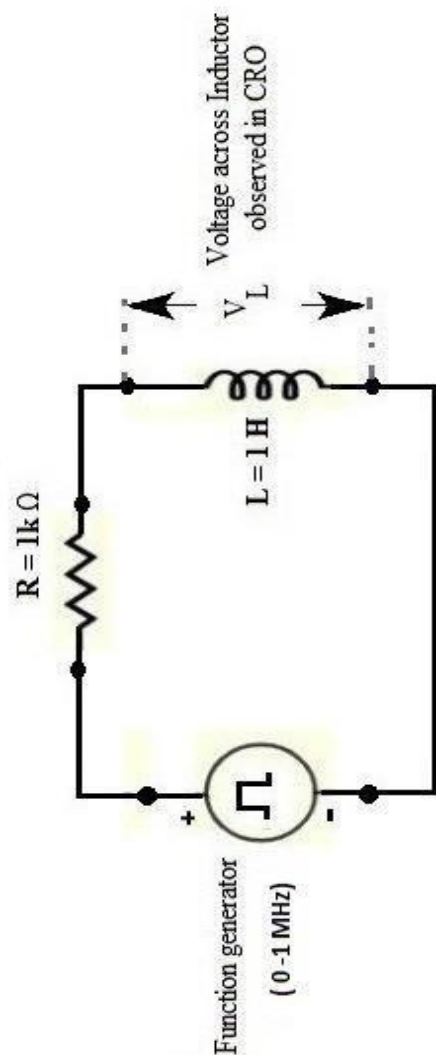
graph 1 : Input voltage waveform to RL series circuit



graph 2 : Transient Response across inductor (L)



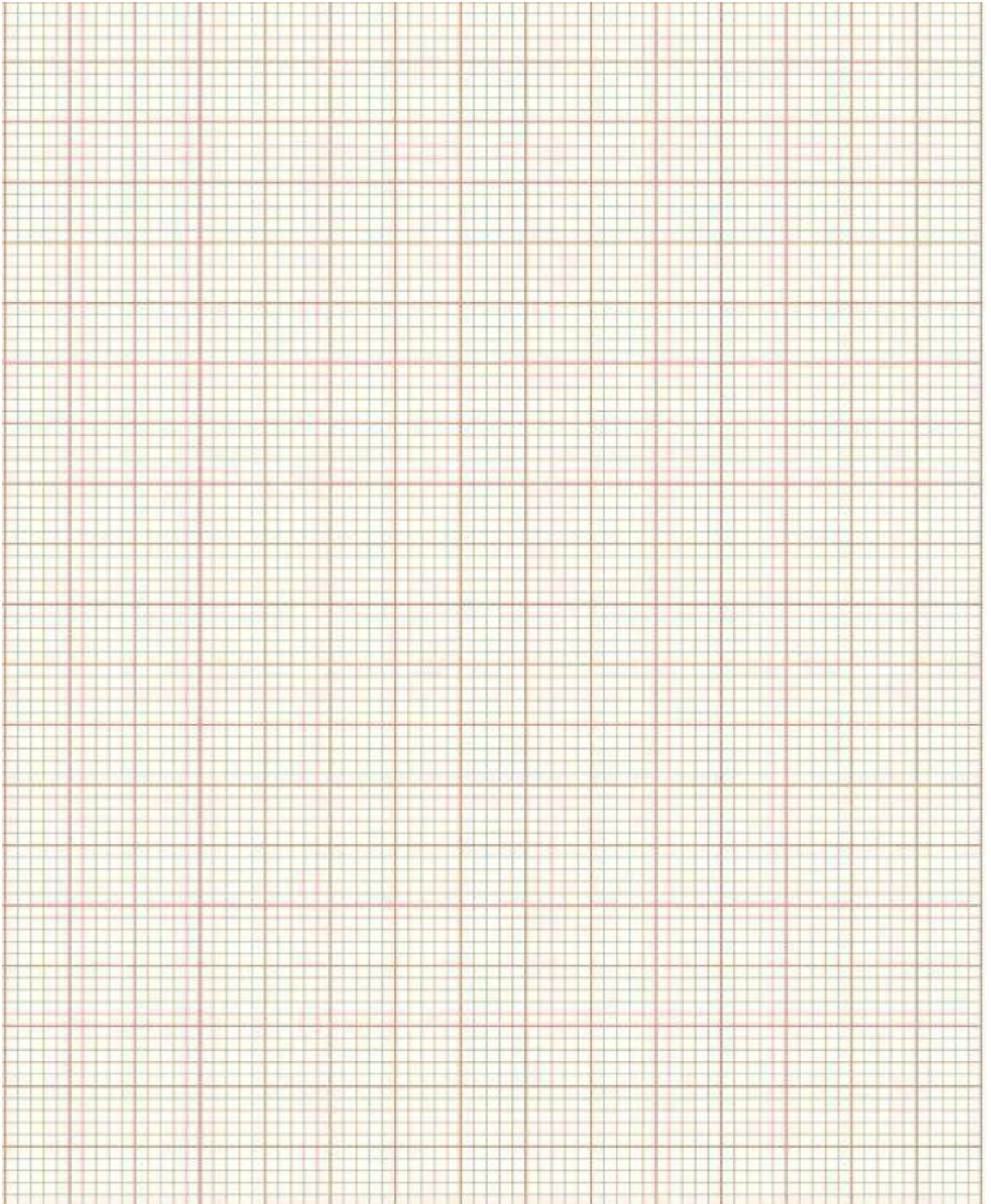
circuit 1: Function Generator acting as DC source



circuit 2 : Series RL circuit excited by DC source

Observation Table:

S. No	Time (m sec)	Voltage(V_L) Theoretical Calculation	Voltage(V_L) from Graph

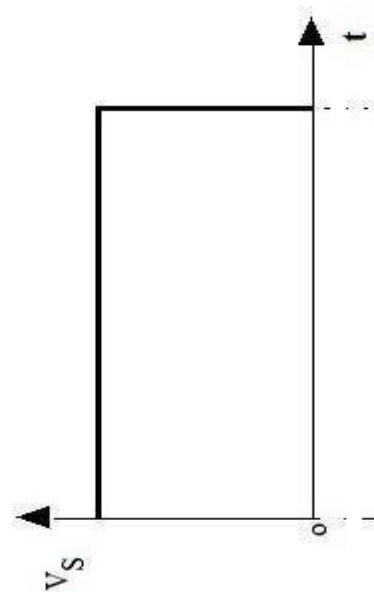
Graph: RL Circuit

Procedure:**Series RC Circuit:**

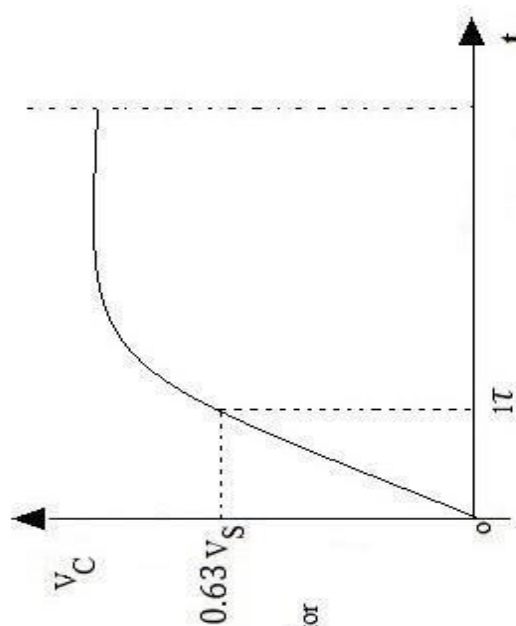
1. Connections are made as per the circuit –1.
2. Turn on Function generator and CRO.
3. Adjust CRO tracer on X-axis using ground (GD) button. Keep the switch in DC mode
4. Set the function generator to deliver a square wave:
 - i. Select function button as square wave, select frequency range as 100 Hz and keep frequency knob at maximum position.
 - ii. Keep the Amplitude knob in maximum position.
 - iii. Turn ON and keep DC offset knob in maximum position and adjust the input waveform to X-axis.
5. Set the VOLTS/DIV to 5V and TIME/DIV to 2ms and draw the input waveform in graph.
6. Connections are made as per the circuit –2, Connect CRO probes across Inductor, adjust the output waveform to X-axis and draw in graph.
7. Find the voltage across capacitor (V_C) at different time constants (τ) and verify it with theoretical calculations.

Series RC Circuit:**Theoretical Calculations:** $V_S = \text{____} V$ Expression of Voltage across Capacitor: $V_C = V_S(1 - e^{\frac{-t}{\tau}})$ Time constant: $\tau = RC =$ Calculate V_C (i) $t = 1 \tau$; $V_C =$ (ii) $t = 2 \tau$; $V_C =$ (iii) $t = 3 \tau$; $V_C =$ (iv) $t = 4 \tau$; $V_C =$ (v) $t = 5 \tau$; $V_C =$

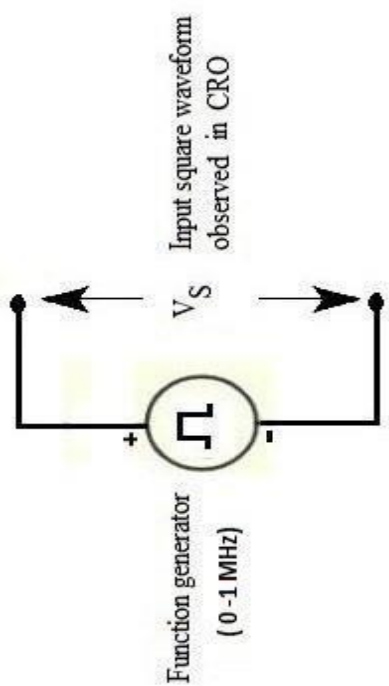
Circuit diagram 3.2: Series RC Circuit:



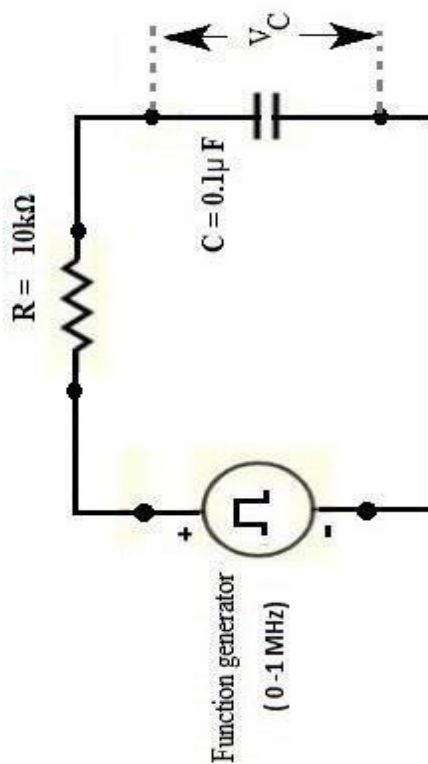
Graph 3 : input voltage waveform to RC series circuit



Graph 4 : Transient Response across Capacitor (C)



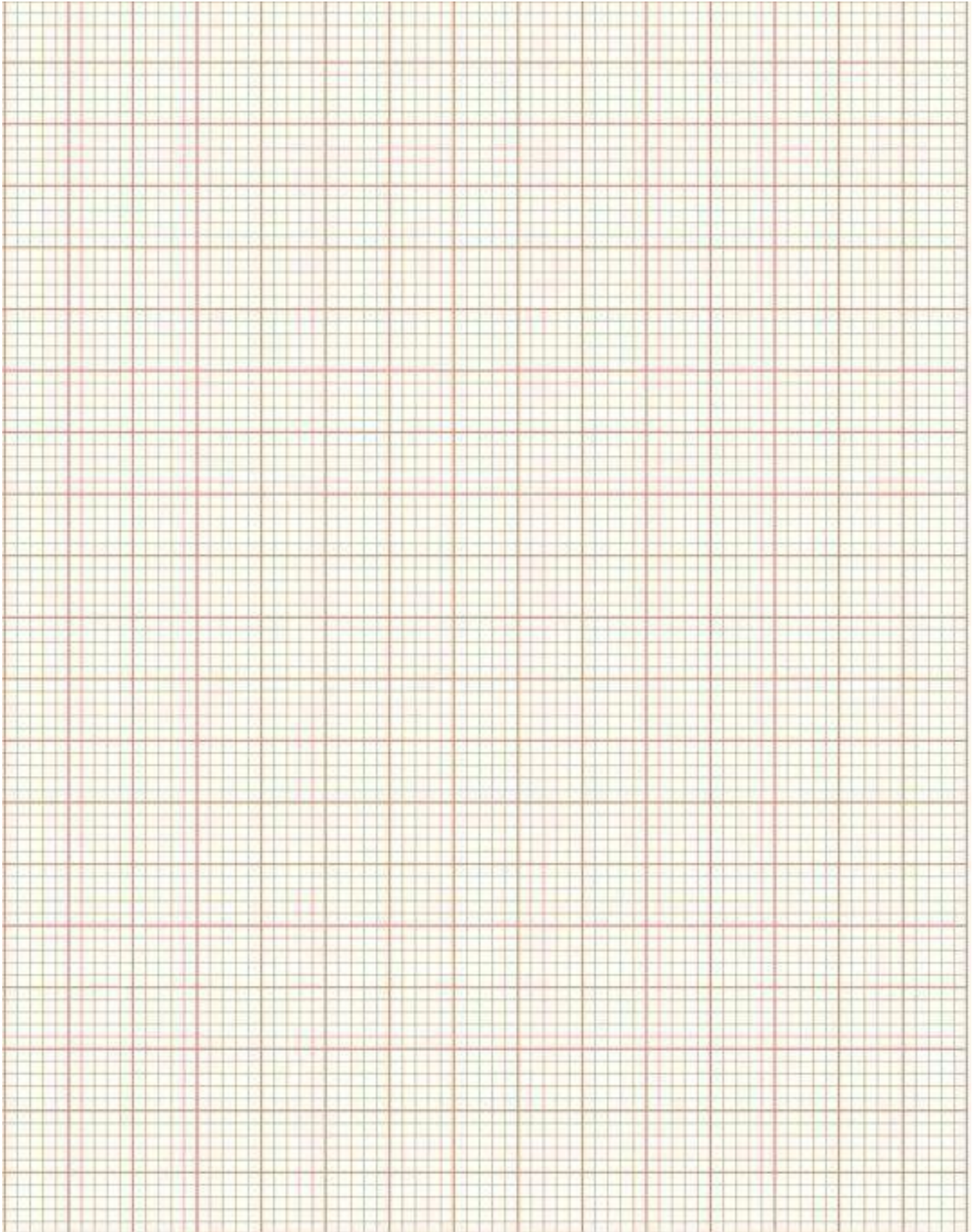
Circuit 3 : Function Generator acting as DC source



Circuit 4 : Series RC circuit excited by DC source

Observation Table:

S. No	Time (m sec)	Voltage(V_C) Theoretical Calculation	Voltage(V_C) from Graph

Graph: RC Circuit

Result:**Discussion Questions:**

1. Define time constant of RL circuit.
2. Why transient occurs in electric circuits?
3. Define time constant of RC circuit.
4. How Inductor and capacitor will behave under initial conditions?
5. How Inductor and capacitor will behave under final conditions?
6. What is function Generator?
7. What is CRO?

Outcome:

By completing this experiment student would be able to know the concept of Transients and able to calculate the transient response of R, L and, C circuits.

Experiment No: _____

Date: _____

RESONANCE IN SERIES 'RLC' CIRCUIT

Objective: To obtain resonance condition in series RLC circuit by varying frequency.

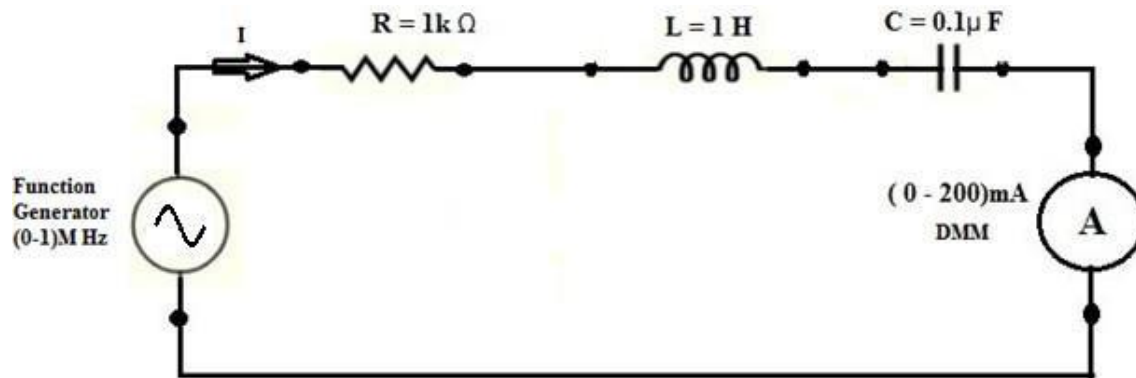
Apparatus:

S. No	Name of the Equipment	Range	Type	Quantity
1	Function Generator	0 - 1 MHz	Digital	1
2	Digital Multi Meter	(0-200) mA	Digital	1
3	Decade Inductance Box	1H	-	1
4	Resistor	1K Ω	Carbon composition or film	1
5	Capacitor	0.1 μF	Film capacitors	1
6	Bread Board	-	-	1
7	Connecting Wires	-	-	as per need

Theory:

Resonance is one of the most important and general phenomenon in almost all branches of science and technology. The resonance can be very useful in devices as oscillators in radio transmitters or electronic watches. More often, however, they can be harmful by generating unwanted wide deflections of mechanical systems (bridges, airplane wings, etc.) or oscillations of voltage and current. Controlling or preventing unwanted resonance is an important aspect of engineering design. Since every electrical component, or even a connecting wire, has some capacitance and inductance there is a potential for resonance in every circuit. That is why design of high frequency circuits is so much more difficult as even small inductance and capacitance of connections plays a role. In this set of experiments you will explore resonance in a series RLC circuit, which have the resonance in relatively low and easy to handle frequency range.

An AC circuit is said to be in Resonance when the applied voltage and current are in phase. Resonance circuits are formed by the combination of reactive elements connected in either series or parallel. In a series RLC circuit there becomes a frequency point where the inductive reactance of the inductor becomes equal in value to the capacitive reactance of the capacitor. Condition of Resonance is $X_L = X_C$. The point at which this occurs is called the Resonant Frequency point(f_r) of the circuit

Circuit Diagram:**Theoretical Calculations:**

Resonance Condition : $X_L = X_C$

$$2\pi fL = \frac{1}{2\pi fC}$$

$$4\pi^2 f^2 LC = 1$$

$$f^2 = \frac{1}{4\pi^2 LC}$$

$$\text{Resonance Frequency (fr)} = \frac{1}{2\pi\sqrt{LC}} \text{ Hz} =$$

$$\text{Quality Factor (Q)} = \frac{2\pi f_r L}{R}$$

$$\text{Bandwidth (B.W)} = \frac{f_r}{Q} \text{ Hz} =$$

Procedure:

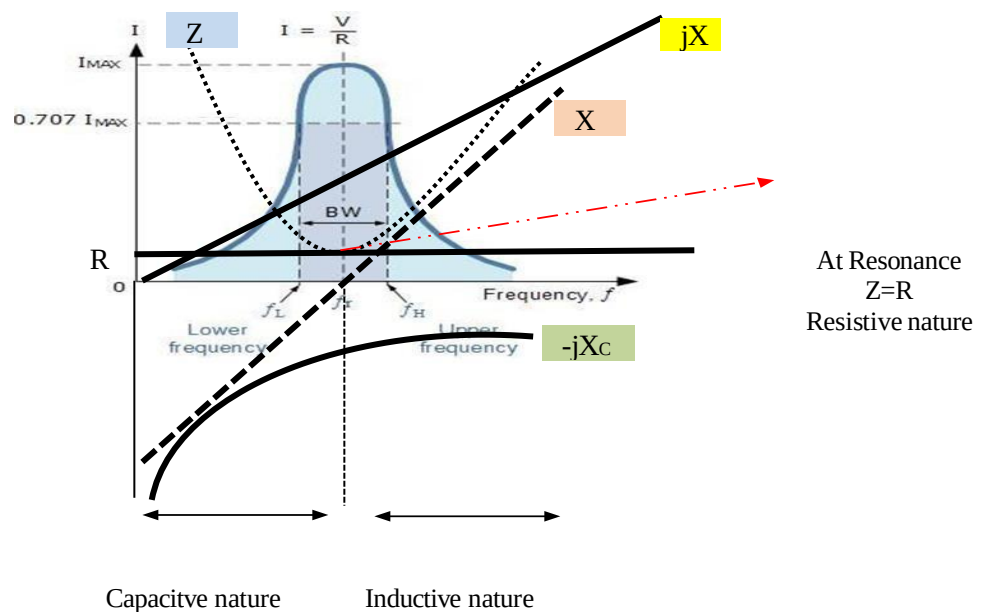
1. Connections are made as per circuit diagram.
2. Set DMM setting to act as Ammeter of range (0-200)mA.
3. Switch on supply and turn on Function generator and DMM.
4. Set the function generator to deliver a sine wave:
 - i. Push in the button to select sine wave and Range.
 - ii. Keep the magnitude knob to maximum position
 - iii. Keep the DC offset knob in off position
5. Frequency knob is slowly varied and for every 100Hz corresponding ammeter reading is noted down up to 1 KHz frequency.
6. After taking reading turn off DMM, function generator and switch off supply.

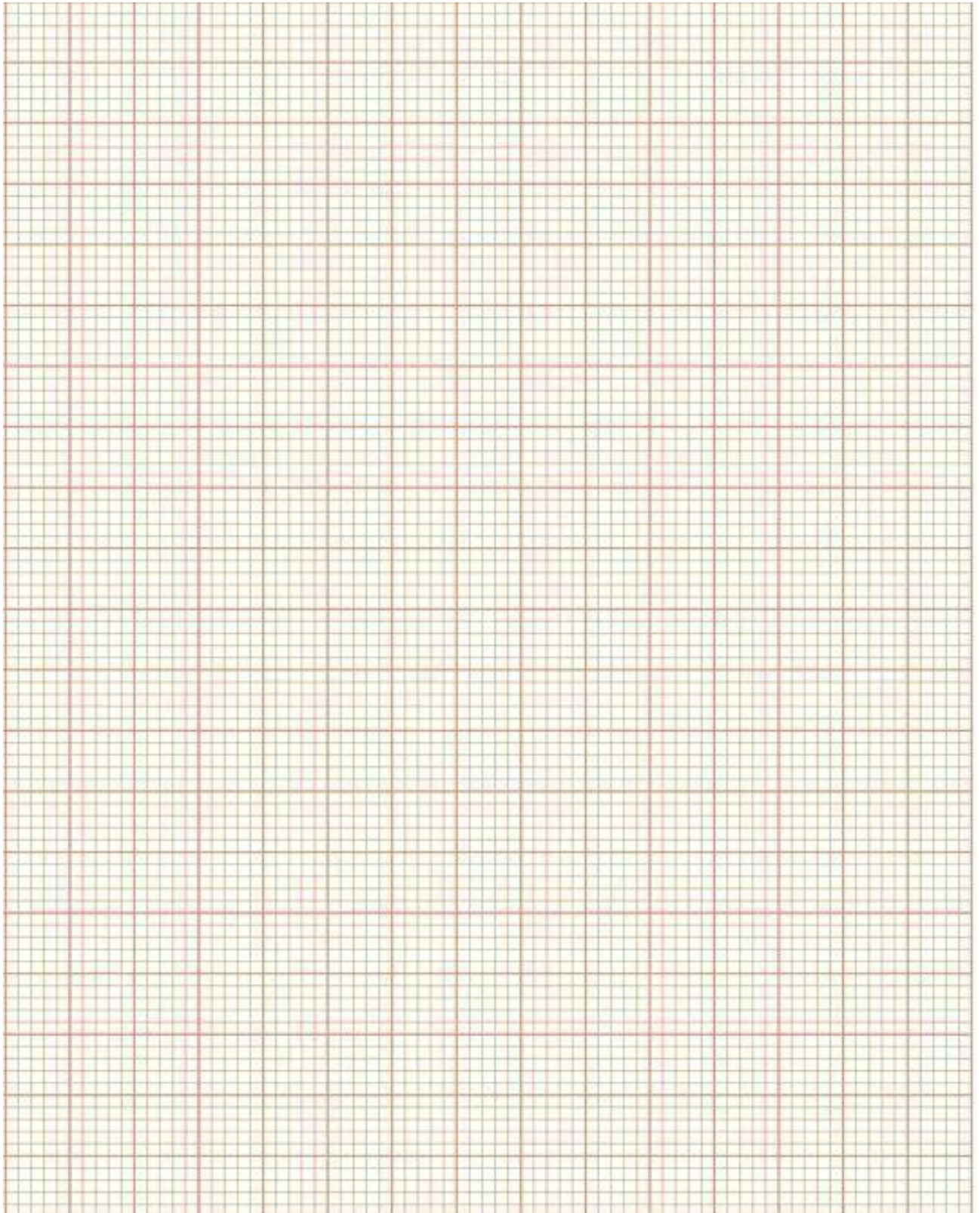
7. A graph is drawn between frequency f and current. Resonant frequency (f_r) and Half power frequencies (f_1, f_2) are marked on the graph by drawing a line parallel to x-axis which represents V_{rms} .
8. Bandwidth and quality factor are found from the graph.
9. Compare resonance frequency theoretically and practically.

Observation Table:

S. No	Frequency (Hz)	Current (mA)	$X_L (\Omega) = 2 \pi f L$ (Inductive Reactance)	$X_C (\Omega) = 1/(2 \pi f C)$ (Capacitive Reactance)	$X = X_L - X_C$ (Net Reactance)	Impedance (Z) $\sqrt{R^2 + (X_L - X_C)^2}$
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						

Model Graph:



Graph:

Result:**Discussion Questions:**

1. Define resonance in electrical circuits and write formula for resonant frequency?
2. What is the condition for resonance for an RLC series circuit?
3. Define Q Factor.
4. Define Bandwidth.
5. What are affects in RLC series circuit at resonance condition?

Outcome:

By completing this experiment student would be able to know the concept of Resonance and able to calculate the resonance of Series RLC circuit.

.....ΩφσδτΠΣ

Cycle - II

.....ΩφσδτΠΣ

Experiment No: _____

Date: _____

CALCULATIONS AND VERIFICATION OF IMPEDANCE AND CURRENT OF 'RLC' SERIES CIRCUIT

Objective:

To calculate Impedance and current of series RLC with R, L and C varying respectively.

Apparatus:

S.No	Name of the Equipment	Range	Type	Quantity
1	Single phase Variac	(230V/0-270V),10A		1
2	Ammeter	(0-2.5)A	MI	1
3	Voltmeter	(0-150)V	MI	1
4	Wattmeter	75V/150/230V,5/10A	LPF	1
5	Rheostat	100Ω/2A	Wire wound	1
6	Inductive Load	(25/50/100)mH/5A	Core	1
7	Capacitive Load	76 μF, 230V/5A		1
8	Connecting wires			As per requirement

Theory:

Impedance is the combination of resistance and reactance. When a circuit is excited by ac source impedance is the total resistance. It is a complex number where the real part is resistance and imaginary part is reactance. It is represented by 'Z' and measured in ohms (Ω). Impedance has both magnitude and phase.

$$Z = V / I$$

Resistance is the property of material which opposes the flow of current and energy is wasted in the form of heat.

$$= /$$

Reactance that occurs in an inductor is known as inductive reactance. When inductive reactance is present, energy is stored in the form of a changing magnetic field, and the current waveform lags the voltage waveform by 90 degrees. Inductive reactance is caused by devices in which wire is wound circularly such as coils (including line reactors), chokes, and transformers.

$$X_L = 2\pi fL$$

Reactance that occurs in a capacitor is known as capacitive reactance. Capacitive reactance stores energy in the form of a changing electrical field and causes current to lead voltage by 90 degrees. Capacitance is created when two conducting plates are placed parallel to one another with a small distance between them, filled with a dielectric material (insulator).

$$X_C = \frac{1}{2\pi fC}$$

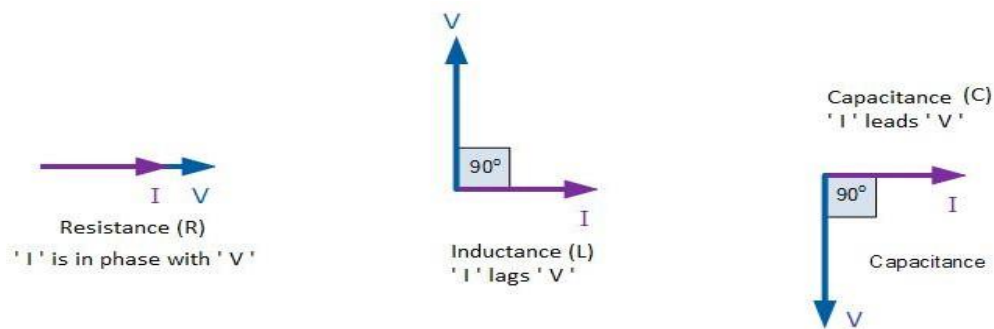


Fig 4.1 . Phasor diagram of V and I in Resistance, Inductance and Capacitance elements.

Calculate the total impedance (Z):

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = V/I.$$

Where R is resistance in ohms.

X_L is inductive reactance in ohms.

X_C is capacitive reactance in ohms.

Total current $I = V/Z$.

The phase angle, which indicates whether the circuit is more inductive or capacitive, is found using:

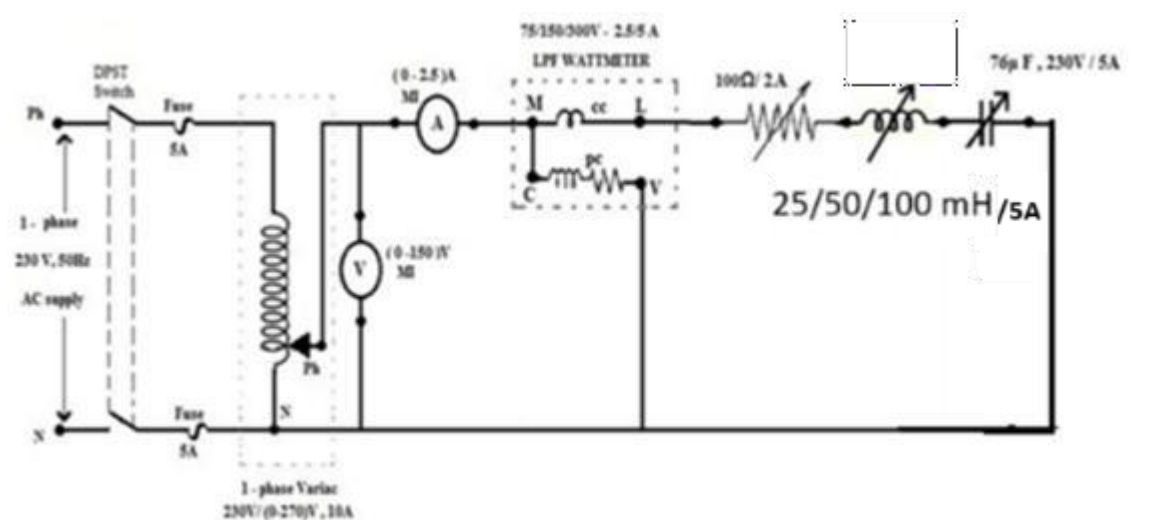
$$\phi = \tan^{-1} [(X_L - X_C) / R]$$

A positive angle indicates an inductive circuit ($X_L > X_C$).

A negative angle indicates a capacitive circuit ($X_L < X_C$).

An angle of zero indicates $X_L = X_C$.

Circuit Diagram : RLC series circuit



Observation Table 1 with constant L and C (R varying):

S. NO.	Input Voltage (V) in volts	Line Current (I) in amps	Power (W) in watts	Impedance $Z = V/I \Omega$	Power Factor $\cos\phi = W/VI$

Procedure: RLC circuit with constant L and C and R varying:

1. All connections are made as per circuit diagram.
2. Make sure Variac is at minimum position, rheostat is kept at minimum position.
3. Switch on DPST and slowly vary Variac and apply 100V by observing voltmeter.
4. Then resistance is varied step by step by keeping L and C fixed.
5. The corresponding ammeter, voltmeter and wattmeter readings are noted down.
6. set Variac to minimum position and switch off DPST.

Observation Table 2 with constant R and C (L varying):

S. NO.	Input Voltage (V) in volts	Line Current (I) in amps	Power (W) in watts	Impedance $Z=V/I \Omega$	Power Factor $\cos \phi =W/VI$

Procedure: RLC circuit with constant R and C (L varying):

1. All connections are made as per circuit diagram.
2. Make sure Variac is at minimum position, rheostat is kept at maximum position.
3. Switch on DPST and slowly vary Variac and apply 100V by observing voltmeter.
4. Now inductance is varied step by step.
5. The corresponding ammeter, voltmeter and wattmeter readings are noted down.
6. Variac set to minimum position and switch off DPST.

Observation Table 2 with constant R and L (C varying):

S. NO.	Input Voltage (V) in volts	Line Current (I) in amps	Power (W) in watts	Impedance $Z=V/I \Omega$	Power Factor $\cos \phi =W/VI$

Procedure: RLC circuit with constant R and L (C varying).

1. All connections are made as per circuit diagram.
2. Make sure Variac is at minimum position; rheostat is kept at maximum position.
3. Switch on DPST and slowly vary Variac and apply 100V by observing voltmeter.
4. Then capacitance is varied step by step.

5. The corresponding ammeter, voltmeter and wattmeter readings are noted down.
6. Switch off capacitive load, set Variac to minimum position and switch off DPST.

Result:**Discussion Questions:**

1. Define impedance.
2. Define Admittance
3. Define Power Factor
4. Define Apparent Power and draw power triangle?
5. What is the purpose of Variac?
6. What are the terminals of Wattmeter? Full form of LPF and UPF wattmeter?

Outcome:

By completing this experiment student would be able to know the concept of Impedance and able to calculate the impedance and currents in R, L and, C circuits.

Experiment No _____

Date: _____

MEASUREMENT OF **VOLTAGE, CURRENT AND REAL POWER** IN PRIMARY AND SECONDARY CIRCUITS OF A SINGLE PHASE TRANSFORMER

Objective:

To measure the voltage, current and real power in primary and secondary circuits of a single-phase transformer.

Apparatus:

S. No	Name of the Equipment	Range	Type	Quantity
1	Voltmeters	(0-150)V	MI	1
		(0-300)V	MI	1
2	Ammeters	(0-1)A	MI	1
3	Wattmeter	75/150/300V, 2.5/5A	LPF	1
4	1- ϕ Transformer	2 KVA, 230/115V	Shell Type	1
5	1- ϕ Auto Transformer (Variac)	230V/(0-270)V, 10A	-	1
6	Connecting Wires	--	-	As per need

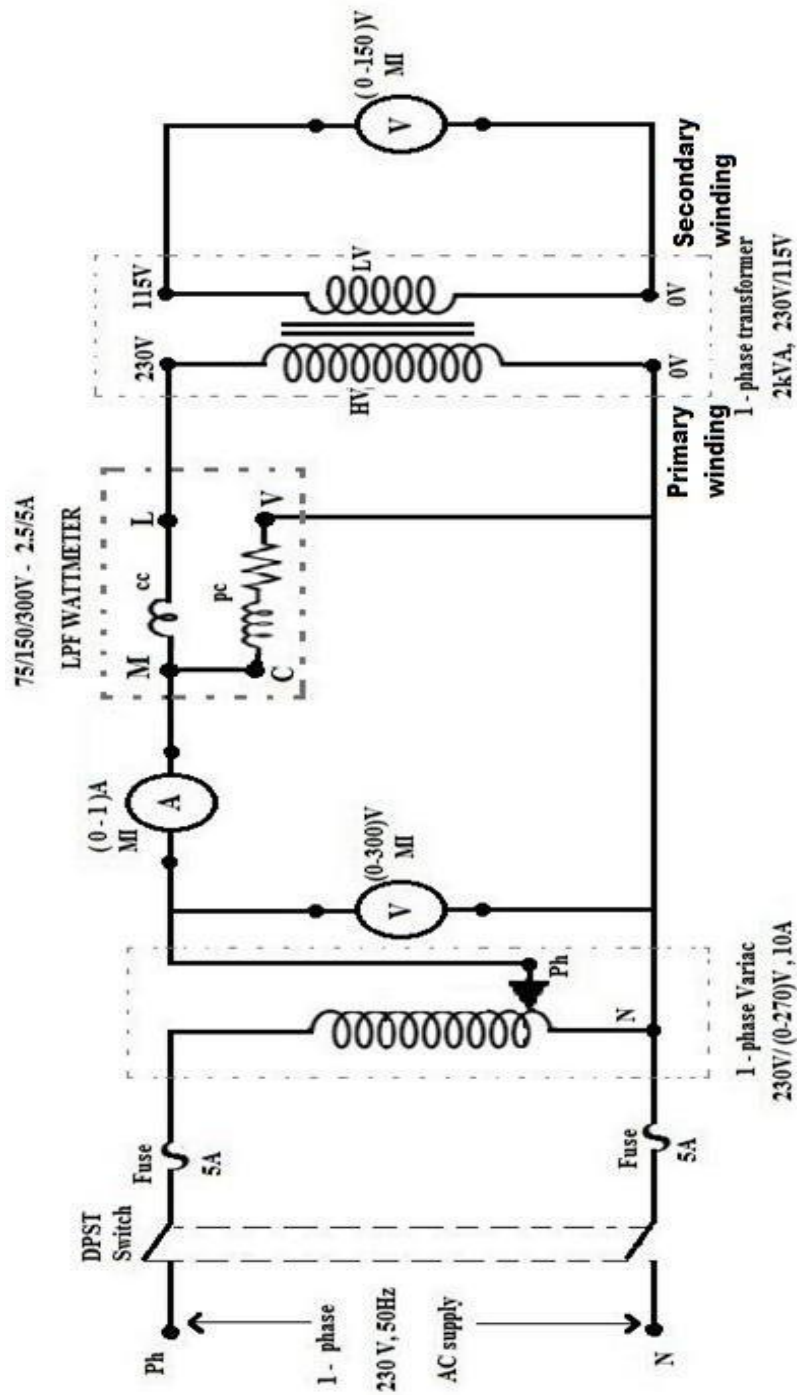
Name plate details:

S. No	Specification	Transformer
1	Rating	
2	Primary Voltage	
3	Primary Current	
4	Secondary Voltage	
5	Secondary Current	

Theory:

Transformer is electromagnetic static electrical equipment (with no moving parts). It consists of a magnetic iron core and copper winding. The transformer is high-efficiency equipment, and its losses are very low because there isn't any mechanical or friction losses. Transformers are used to step up and step down voltage maintaining frequency constant

Circuit diagram:-HV as primary side:



S.No	Primary voltage(V)	Primary current (A)	Real power(W)	Secondary voltage(V)
1				

It operates only with alternating current (AC), because the direct current (DC) does not create any electromagnetic induction.

The main parts of a transformer are windings, core, and insulation. The windings should have small resistance value and usually they are made of copper. They are wound around the core and must be isolated from it. Also, the windings turns have to be isolated from each other. The transformer core is made from very thin steel laminations which have high permeability. The laminations have to be thin (between 0.25 mm and 0.5 mm) because of decreasing power losses (known as eddy current losses). To reduce hysteresis losses high permeability material is used.

When an AC supply is given to the primary side of the transformer current will flow in the winding creating a magnetic flux which on interaction with secondary induces an emf in secondary but as per the circuit given secondary is not connected to any load hence the current drawn in the primary is only used to magnetize core of the transformer. Let's say I_0 is the no-load current and it is in the range of 2% - 5% of rated current.

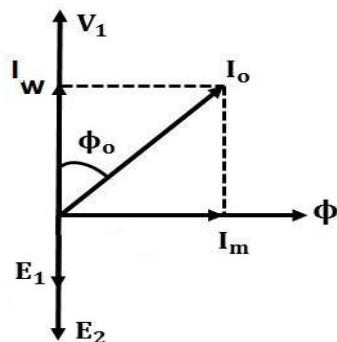


Fig 6.1 no load phasor diagram

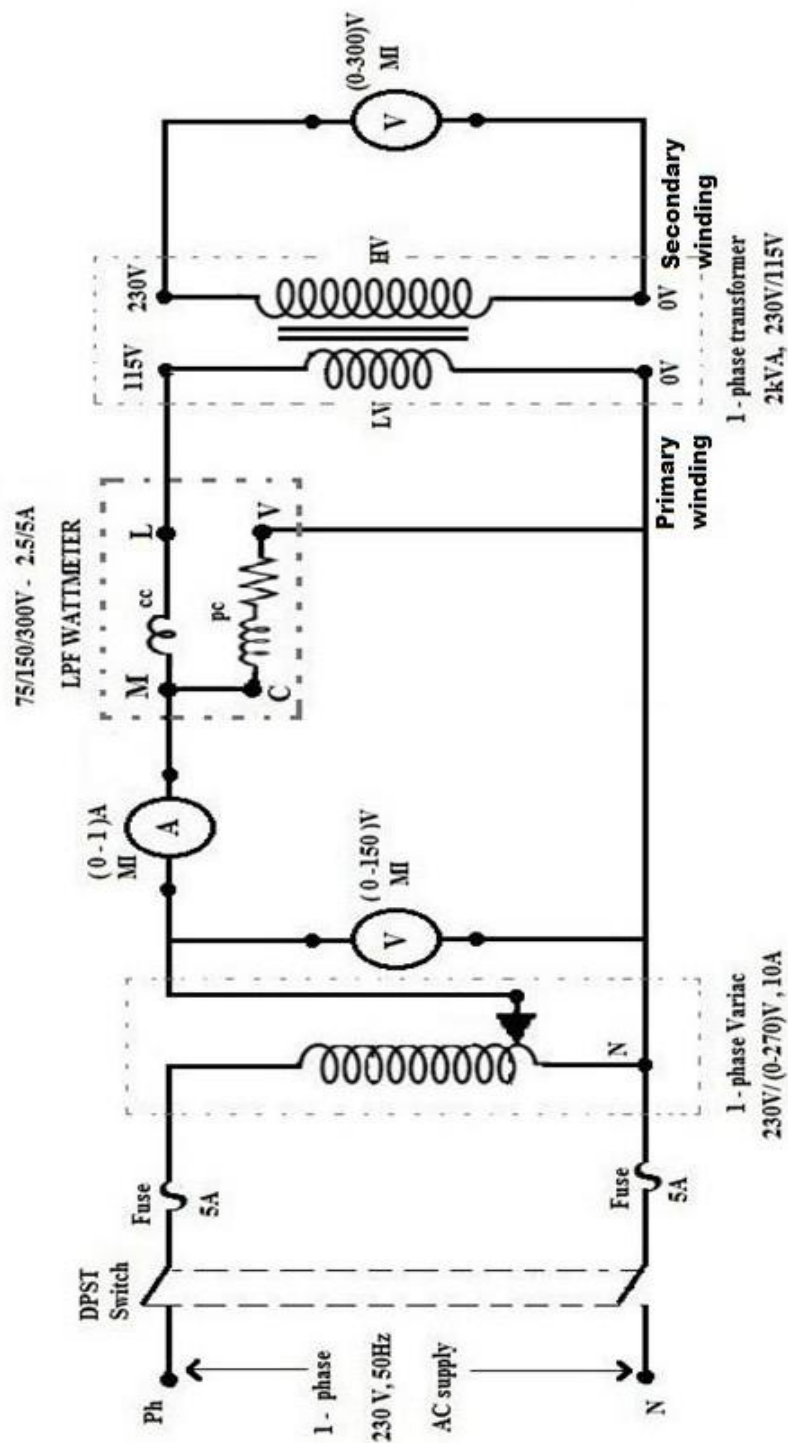
$$= \sqrt{\quad + \quad}$$

$$\Phi = \tan^{-1} \left(- \right)$$

Procedure:

1. Connections are made as per circuit diagram
2. Make sure that the Variac is in minimum position.
3. Supply is switched on using DPST and rated voltage is applied to the HV side by varying the Autotransformer.
4. Note down the readings of currents, voltages and real power using Ammeter, Voltmeter & Wattmeter respectively.
5. Adjust the Variac to minimum position and switch off DPST.

Circuit diagram: - LV as primary :



S. No	Primary voltage(V)	Primary current (A)	Real power(W)	Secondary voltage(V)
1				

Procedure:

1. Connections are made as per circuit diagram
2. Make sure that the Variac is in minimum position.
3. Supply is switched on and rated voltage is applied to the LV side by varying the Auto transformer.
4. Note down the readings of currents, voltages and real power using Ammeter, Voltmeter & Wattmeter respectively.
5. Adjust the Variac to minimum position and switch off DPST.

Result:**Discussion Questions:**

1. What is a Transformer and transformation ratio (k)?
2. Why rating of transformer is specified in kVA?
3. What happens when DC supply is given to the transformer?
4. What are the various losses that occur in Transformer?

Outcomes:

1. Students will be able to understand the function of STEP-UP and STEP-DOWN transformer.
2. Student will be able to calculate the Real power in Transformer.

Experiment No _____

Date: _____

LOAD TEST ON A SINGLE-PHASE TRANSFORMER

Objective:

To conduct Load test on the given 1- Φ Transformer and to calculate its efficiency and Regulation.

Apparatus:

Name plate details:

S. No	Name of the Equipment	Range	Type	Quantity
1	Voltmeters	(0-150)V	MI	1
		(0-300)V		1
2	Ammeter	(0-10)A	MI	1
3	Wattmeter	75/150/300V, 5/10A	UPF	1
4	1- ϕ Transformer	2KVA, 230/115V	Shell Type	1
5	1- ϕ Auto Transformer (Variac)	230V/(0-270)V, 10A	----	1
6	Resistive load bank	5kW , 230V	Variable	1
7	Connecting Wires	---	---	As per need

S. No	Specification	Transformer
1	Rating	
2	Primary Voltage	
3	Primary Current	
4	Secondary Voltage	
5	Secondary Current	

Theory:

Transformer is electromagnetic static electrical equipment (with no moving parts). It consists of a magnetic iron core and copper winding. The transformer is high-efficiency equipment, and its losses are very low because there isn't any mechanical or friction losses. Transformers are used to step up and step down voltage maintaining frequency constant It

operates only with alternating current (AC), because the direct current (DC) does not create any electromagnetic induction.

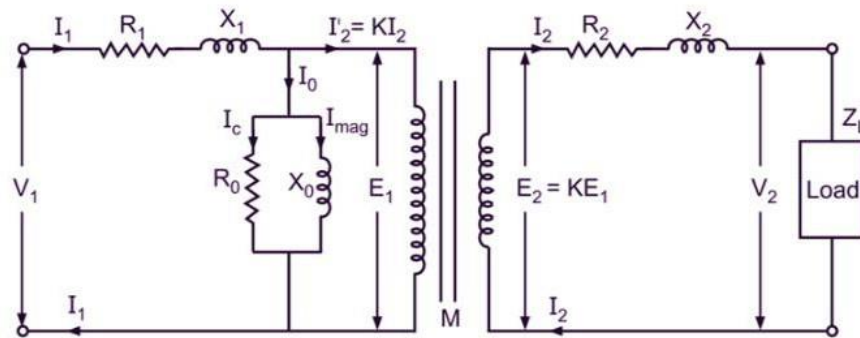
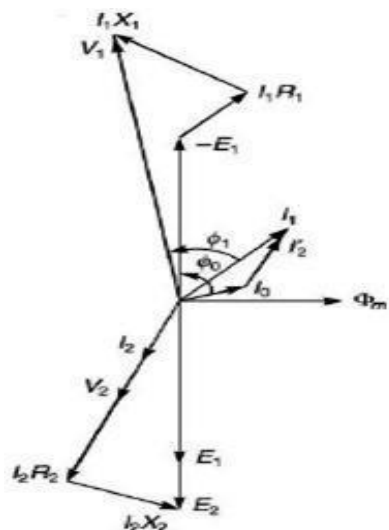


Fig 7.1 Equivalent circuit of a single phase transformer

When load is connected to the secondary side of the transformer, current I_2 drawn from in the secondary to serve loaded condition. To compensate load variation, current I'_2 in addition to the I_0 has to be provided by input voltage source V_1 . below figure shows the phasor diagram when transformer is connected to a resistive load



$$= +'$$

$$1 = 1 + 1Z_1$$

$$= + Z_2$$

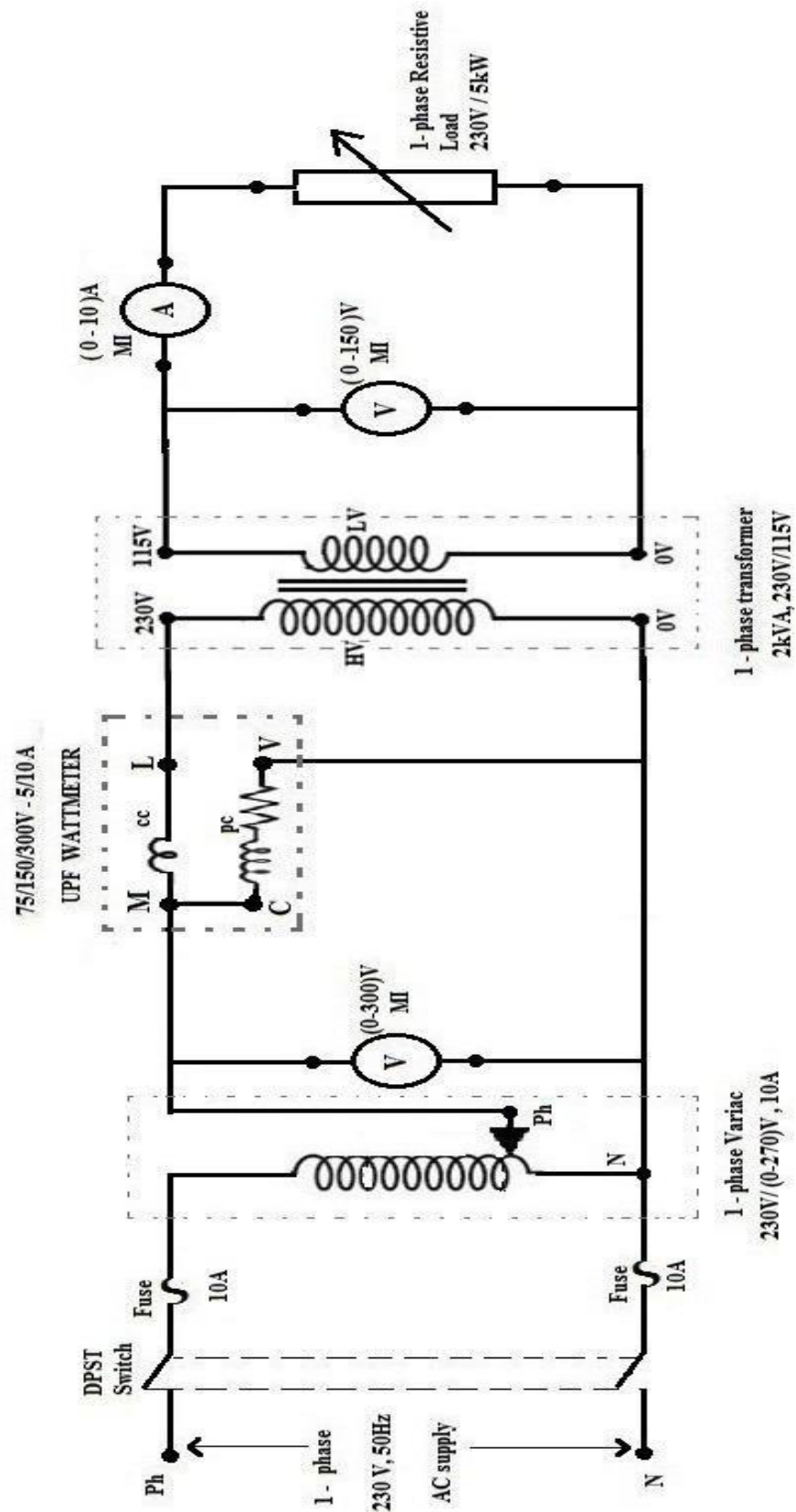
Regulation indicates how much is the variation of transformer secondary voltage from no load to full load. This change is basically because of the winding resistance and reactance.

$$\% \text{ Regulation} = \frac{2 d-2 d_x}{2 d} \quad 100$$

Efficiency of transformer is the ratio of the output power to the input power in a transformer.

$$\% \text{ Efficiency} = \frac{wu}{pu} \times 100$$

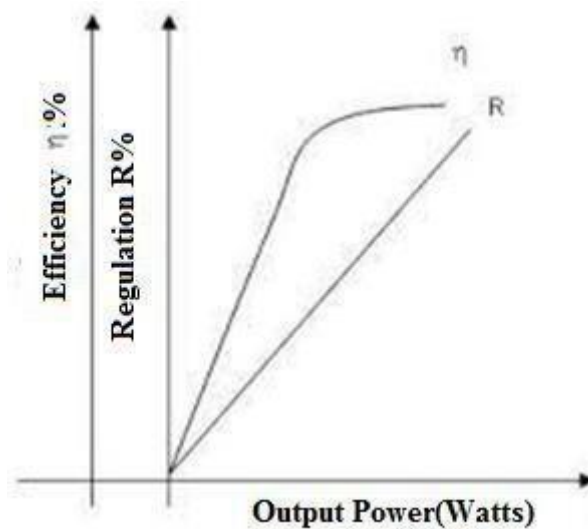
Circuit diagram :



Procedure:

1. Connections are made as per the circuit diagram.
2. Make sure that the Variac is in minimum position and resistive load is off.
3. Switch on the DPST.
4. By varying the Variac, rated voltage is applied to the input side of the transformer and should be maintained constant throughout the experiment.
5. Before applying any load note down readings of voltmeter, ammeter and wattmeter under no load operation.
6. By varying the load in steps, readings of ammeter, voltmeters, and wattmeter are noted down in each step.
7. After readings are noted down slowly vary Variac and set it to minimum position and switch off DPST.
8. Efficiency and Regulations are calculated in each step and tabulated.
9. Graphs are drawn Output Vs Efficiency.

Model Graph:



Note:

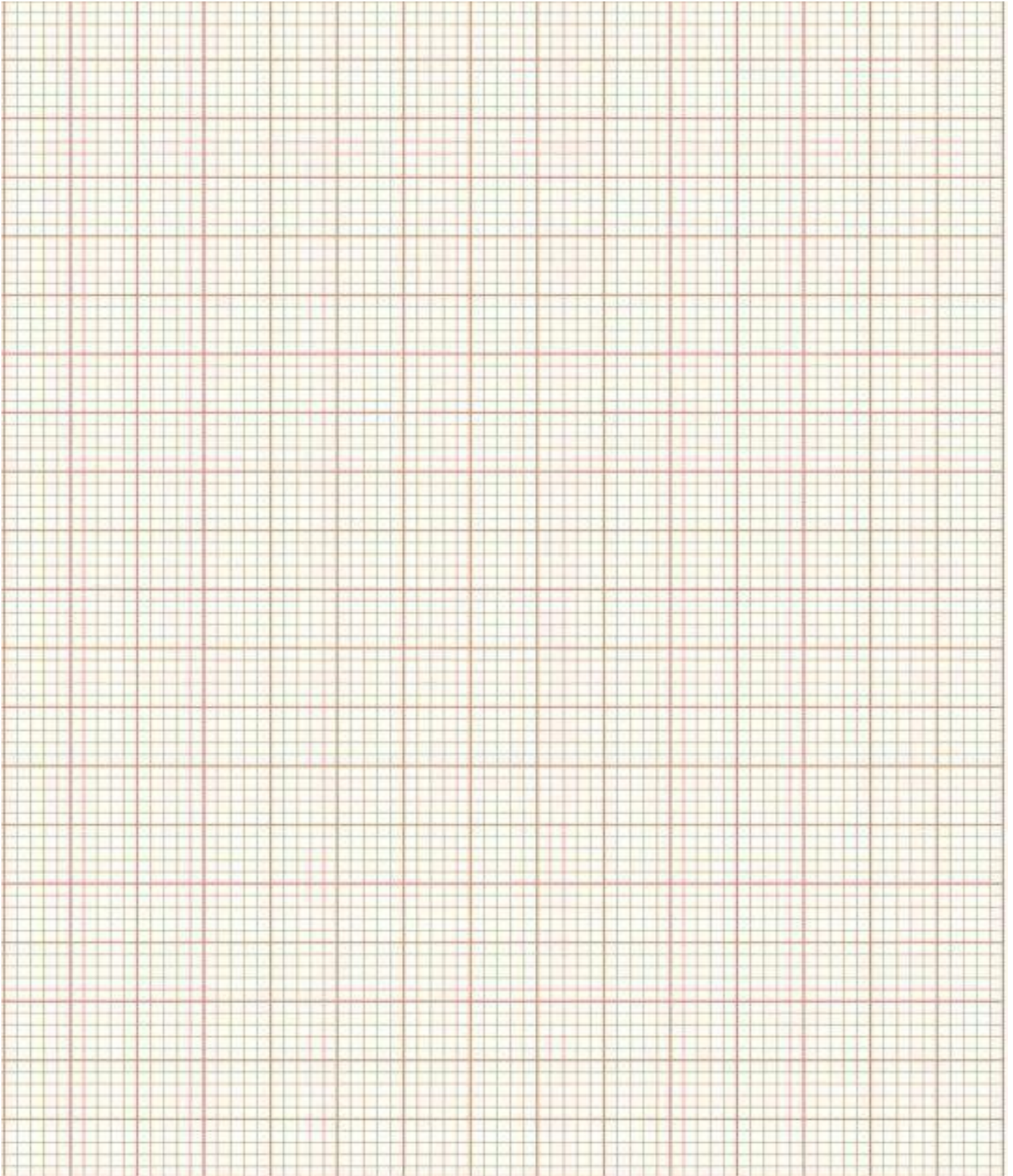
- When a winding of transformer is connected to AC supply it is called Primary winding and if connected to load it is secondary winding.
- HV and LV winding classification can be identified by number of turns of winding.

Observation table:

S. No	V ₁ Volts	Input power W Watts	V ₂ Volts	I ₂ Amp	Output Power = V ₂ x I ₂	Efficiency= $\frac{\text{Output Power}}{\text{Input Power}} \times 100$	Regulation= $\frac{V_{no-load} - V_{load}}{V_{load}} \times 100$

Calculations:

Graph :



Result :**Discussion questions:**

1. What is the function of a Transformer?
2. Define voltage regulation of a Transformer?
3. What are the different parts of a Transformer?
4. What is the purpose of Auto transformer (or Dimmer stat)?
5. What is impedance ratio and draw equivalent circuit of transformer with respect to secondary?

Outcomes:

1. Student will be able to understand how practically the transformer is loaded.
2. Student will be able to calculate the regulation and efficiencies at different loads.

Student also will be able to identify the winding on the transformers and from practical data calculations they are able to design the small rating transformers.

Experiment No:_____

Date:_____

PERFORMANCE CHARACTERISTICS OF A DC SHUNT MOTOR

Objective:

To conduct Brake test on a self-excited DC Shunt motor and to draw its performance characteristics.

Apparatus:

S. No	Name of the Equipment	Range	Type	Quantity
1	Voltmeter	(0-300)V	MC	1
2	Ammeter	(0-10)A	MC	1
3	Rheostat	350 Ω /1.7A	wire wound	1
5	Tachometer	(0-9999)RPM	Digital	1
6	Connecting Wires		-	As per need

NAME PLATE DETAILS:

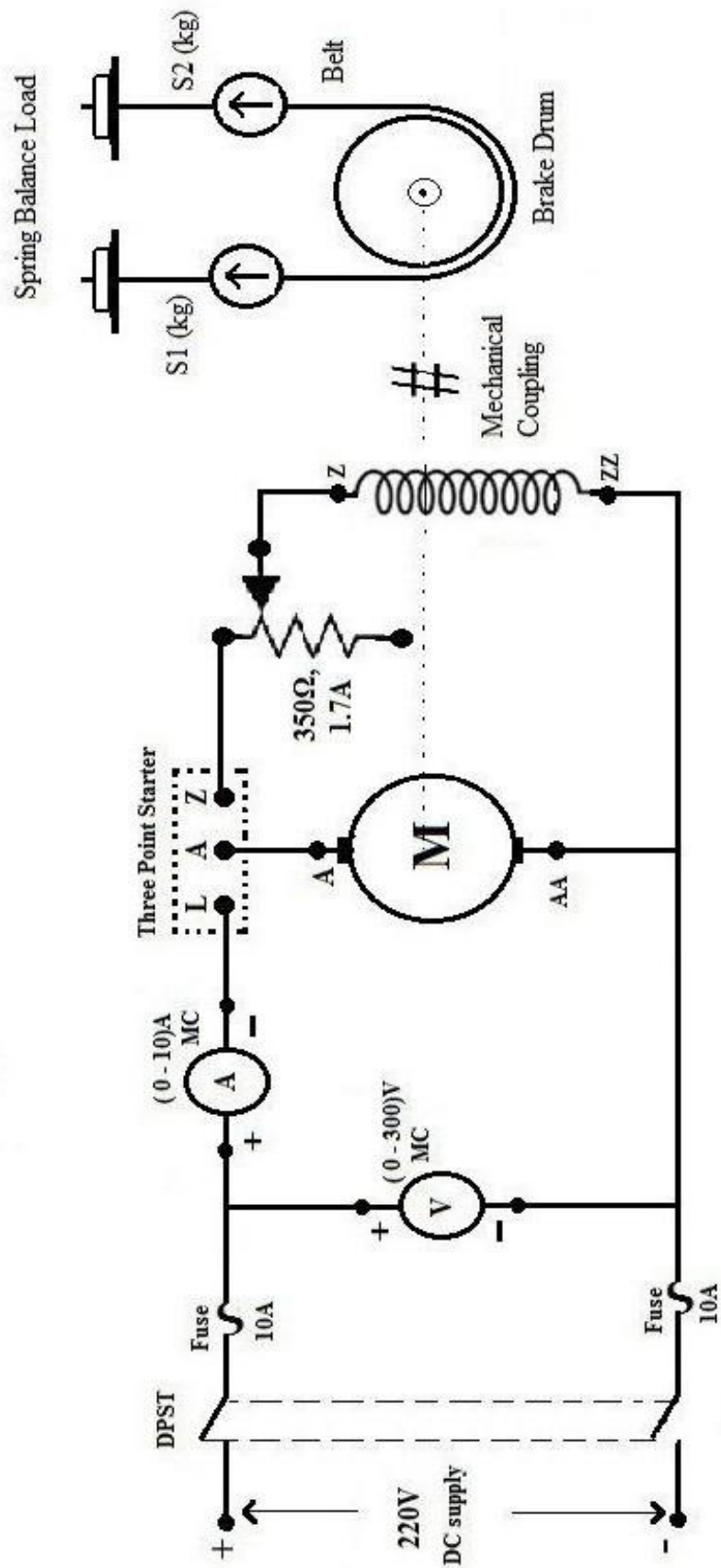
S. No	Specification	Motor
1	Rated Voltage	
2	Rated Current	
3	Speed	
4	Power	
5	Type of Excitation	

Theory:

- It is a simple method of testing low rating DC machines and consists of applying a brake to a water-cooled drum mounted on the motor shaft.
- The four important characteristics curves of a D.C. Shunt Motor, namely, Torque, Speed, Load Current & efficiency, each plotted against the output Power, as shown in the model graph are known as Performance characteristics.
- A belt is wound round the brake drum and its two ends are attached to two spring balances S1 & S2.
- The tension of the belt can be adjusted with the help of swivels.

- The force acting tangentially on the drum is equal to the difference between the readings of the two spring balances.
- The net force, F applied on the brake drum is $9.81(S_1 - S_2)r$ Newton's where , S_1 & S_2 are the readings of Spring balances 1& 2 in Kg.
- Shaft torque T , developed by the motor is $9.81 (S_1 - S_2) R$ Nm where, R is the radius of the pulley in meters & N is the speed in rpm.
- Useful Output Power = $(2 \pi N T) / 60$ Watts
- % Efficiency , = $(\text{Output power} / \text{Input power}) \times 100$
- Input Power = $V I_L$ Watts.
- The size of the motor that can be tested by this method is limited from the consideration of the heat that can be dissipated at the brake drum
- Where the output power exceeds about 2 H.P., or where the test is of long duration, its necessary to use a water cooled brake drum.

Circuit diagram:



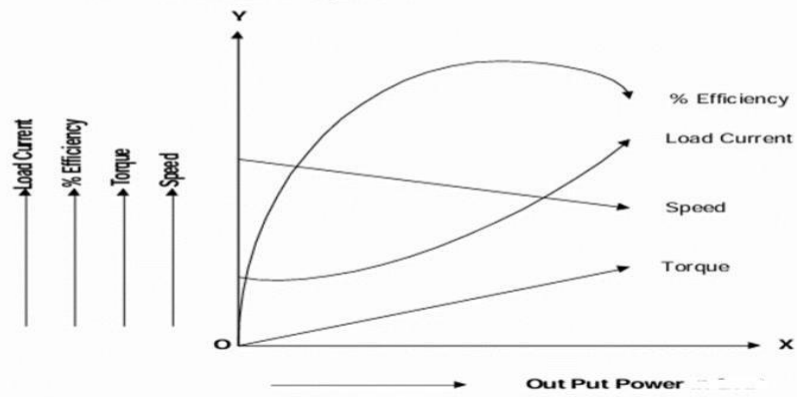
Procedure:

1. Make the connections as per circuit diagram.
2. Keep the field rheostat of the Motor at minimum Resistance position.
3. Check that the belt on the brake drum is free moving before supply is switched on.
4. Turn ON DPST switch and motor is started by slowly moving handle of three point starter.
5. Using tachometer measure speed of motor. If the measured speed is not equal to rated speed of motor slowly adjust the field rheostat so that motor runs at its rated speed.
6. Take Ammeters, Voltmeter, Tachometer and spring balance load readings at rated speed of motor. (at no load condition)
7. Pour water in the brake drum before applying load.
8. By increasing load in steps, Note down the readings of Ammeters, Voltmeter, speed and two spring balance readings (S_1 and S_2) till 75 % of full load current of motor.
9. After completing experiment slowly remove load till the belt is freely moving, rheostat position is adjusted to minimum and turn off DPST.

Model Graph:

Expected Graphs:

- i. % Efficiency Vs Output Power
- ii. Speed Vs Output Power
- iii. Torque Vs Output Power
- iv. Load Current Vs Output Power.



Observation Table:

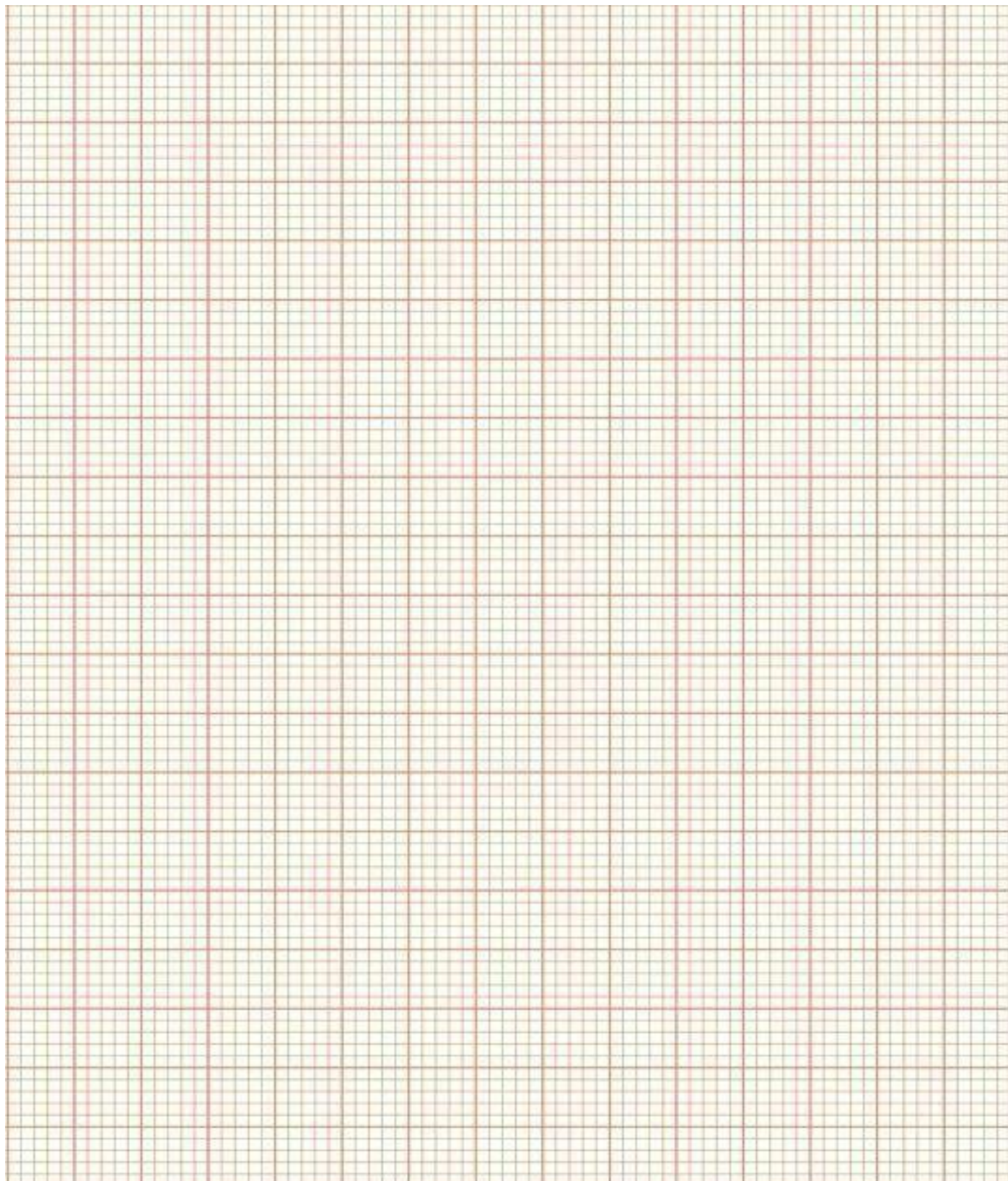
S. NO	Voltage VL (V)	Current IL (A)	Speed N (rpm)	S1 (kg)	S2 (kg)	Input Power VLIL (W)	Torque T (N-m)	Output Power (W) = $2 \pi NT/60$	% Efficiency, $\eta = \text{Output/Input} \times 100$

Calculations:

Radius of the Brake drum (r) = ____m

Torque (T) = $9.81 \times (S1 - S2) \times r$ = ____N-m

Graph:



Result:**Discussion Questions:**

1. What is Torque ?
2. Name the various types of DC Motors.
3. What is the purpose of using three point starter?
4. Write applications of DC shunt and series motor?
5. What is the significance of back emf ?
6. Give the disadvantages of brake test?

Outcomes:

1. Students will be able to find the Torque of self excited DC shunt motor
2. Students will be able to draw the TORQUE-SPEED characteristics of self excited DC shunt motor.

Experiment No: _____

Date: _____

TORQUE – SPEED CHARACTERISTICS OF
THREE PHASE INDUCTION MOTOR

Objective: To draw the Torque-Speed characteristics of a three-phase induction motor.

Apparatus:

S. No	Name of the Equipment	Range	Type	Quantity
1	Ammeter	(0-5)A	MI	1
2	Voltmeter	(0-600)V	MI	1
3	Tachometer	(0-9999) rpm	Digital	1
4	Connecting Wires	-	-	As per need

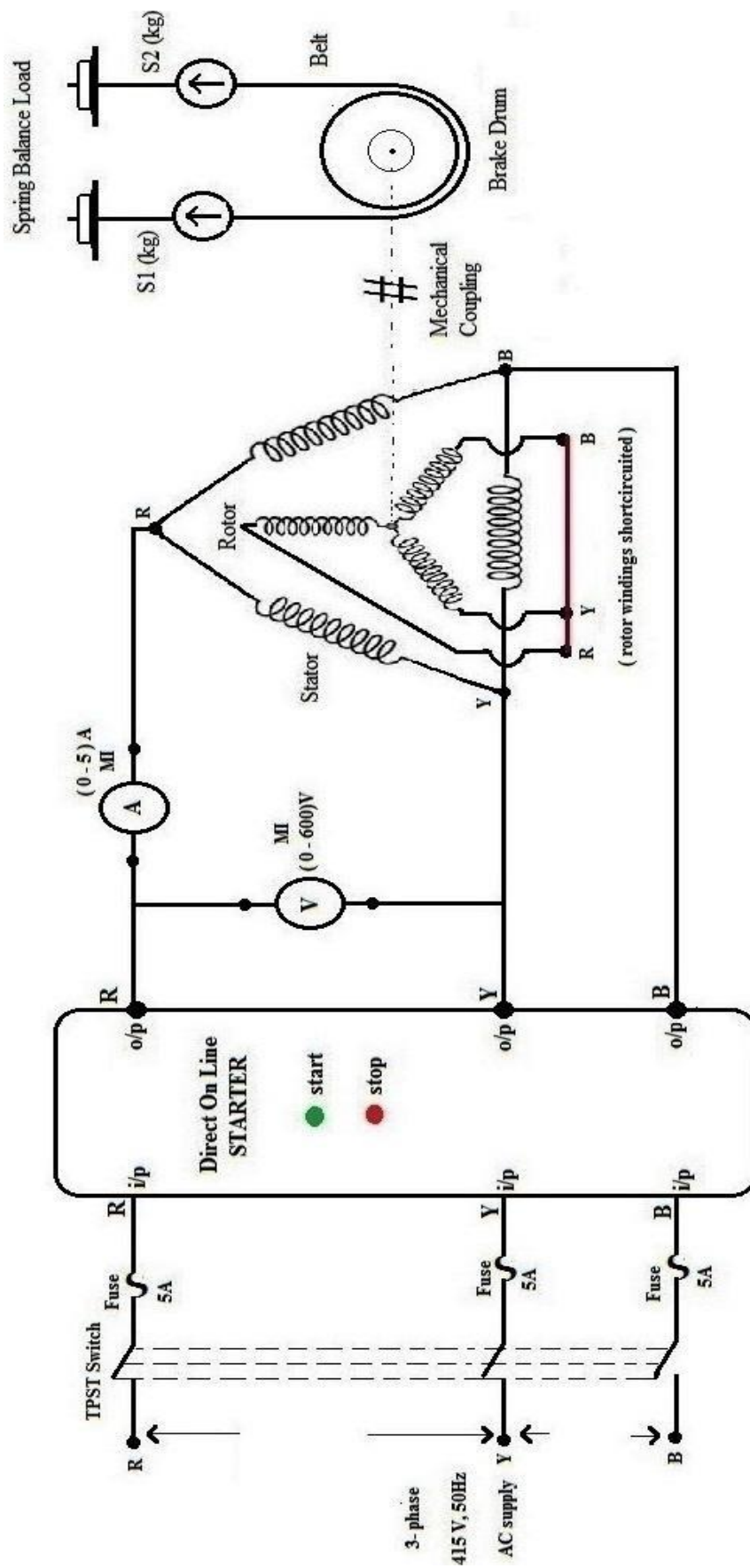
Name Plate Details:

S. No	Specification	Motor
1	Rated Voltage	
2	Rated Current	
3	Speed	
4	Power	
5	Frequency	

Theory:

Slip ring induction motor is a type of induction motor in which the rotor is provided with 3 phase double layer distributed winding consisting of coils as used in alternators. The rotor is wound for as many poles as the number of stator poles and is always wound 3 phase even when the stator is wound two phase. The three phases are starred internally. The other three winding terminals are brought out and connected to three insulated slip rings mounted on the shaft with brushes resting on them. These brushes are further externally connected to a three phase star connected rheostat. This makes possible the introduction of additional resistance in the rotor circuit during the starting period for increasing the starting torque of the motor and for changing its speed torque /current characteristics.

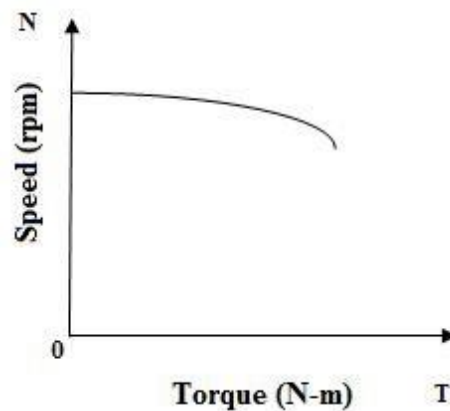
Circuit Diagram:



Procedure:

1. Make the connections as per circuit diagram.
2. Check that belt on the brake drum is free moving before supply is switched on.
3. Switch ON the TPST switch, then start the motor by using DOLstarter by pressing start button
4. Take no load readings of ammeter, voltmeter and speed of the motor.
5. Pour water in the brake drum before applying load.
6. By increasing load in steps, Note down the readings of ammeter, voltmeter, speed and spring balance readings (S1, S2).until it reaches 75% of Full current of motor.
7. Release the load on motor, switch off the DOL starter by pressing Stop button and turn off TPST switch

Expected Graph: Torque Vs Speed



Observation Table:

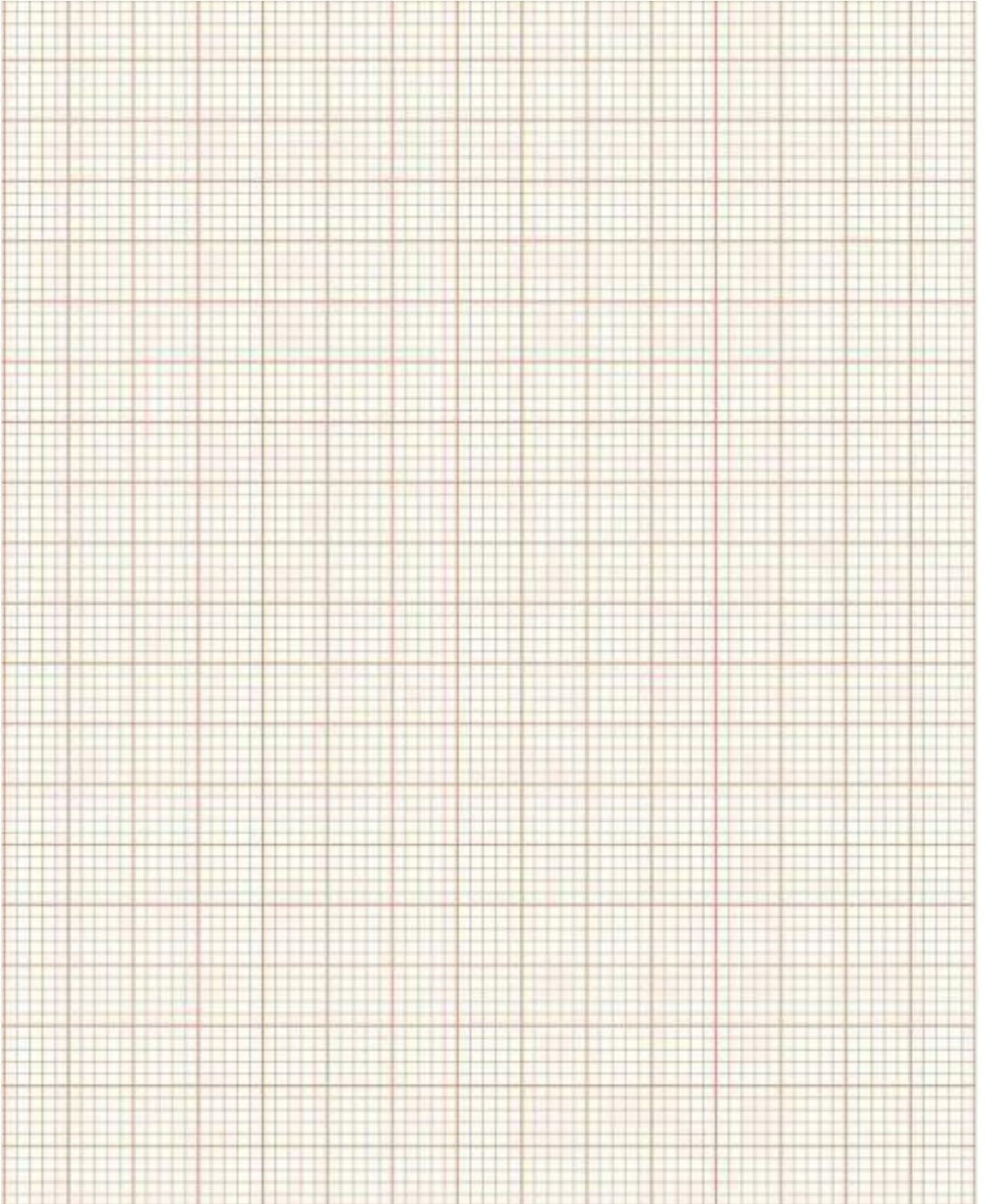
S. No	V _L (V)	I _L (A)	S ₁ (Kg)	S ₂ (Kg)	Speed(N) in RPM	Torque(T) in N-m
1						
2						
3						
4						
5						
6						
7						
8						
9						

Calculations:

Radius of the Brake drum (r) = ____m

Torque (T) = $9.81 \times (S_1 - S_2) \times r$ = ____N-m

Graph:



Result:**Discussion Questions:**

1. Give classification of Three Phase induction motors.
2. What is Rotating Magnetic field (RMF)?
3. What is synchronous speed?
4. What is Slip? Write formulae of slip and rotor frequency?
5. Why do we conduct brake test on Motors?
6. Write some application of 3 phase induction motors

Outcome:

1. Students will be able to find the Torque of Three phase induction motor
2. Students will be able to draw the TORQUE-SPEED characteristics of Three phase induction motor.

.....ΩφσδτΠΣ

Additional EXPERIMENTS

.....ΩφσδτΠΣ

Experiment No: _____

Date: _____

THREE PHASE TRANSFORMER: VERIFICATION OF RELATIONSHIP BETWEEN VOLTAGES AND CURRENTS (STAR-DELTA, DELTA- DELTA, DELTA-STAR, STAR-STAR)

Objective: Three Phase Transformer: Verification of Relationship between Voltages and currents (Star-Delta, Delta-Delta, Delta-star, Star-Star)

Apparatus:

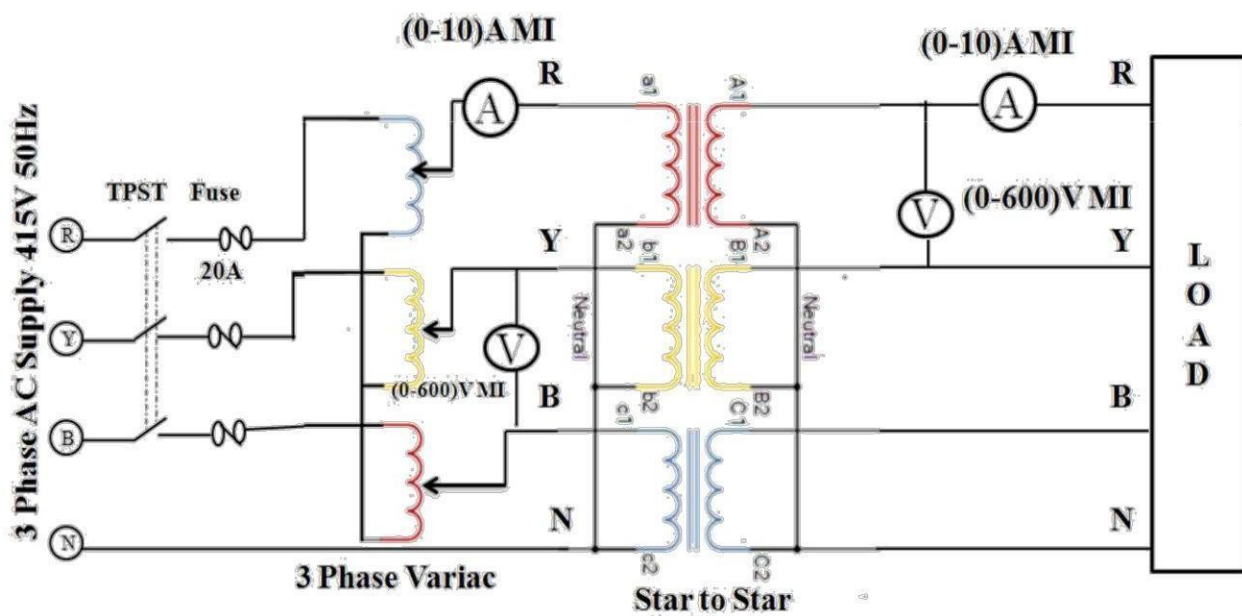
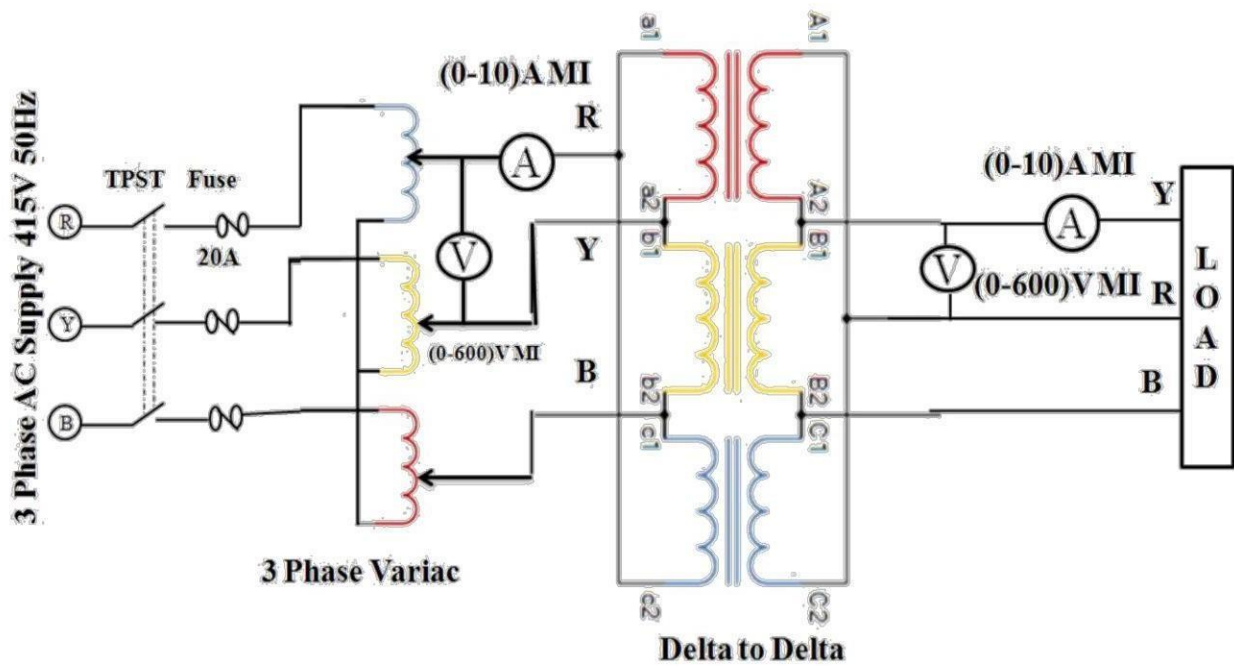
S. No	Name of the Equipment	Range	Type	Qty
1	Voltmeters	(0-600)V	MI	2
2	Ammeters	(0-10)A	MI	2
3	Watt - meters	75/150/300V, 5/10A	LPF	1
4	3- ϕ Transformer	0.6 kVA	Shell Type	1
5	3- ϕ Auto Transformer (Variac)	440V/(0-470)V, 4A	-----	1
6	Connecting Wires	--	-----	As per need

Theory:

A three phase transformer or 3 ϕ transformer can be constructed either by connecting together three single-phase transformers, thereby forming a so-called three phase transformer bank, or by using one pre-assembled and balanced three phase transformer which consists of three pairs of single phase windings mounted onto one single laminated core. The advantages of building a single three phase transformer is that for the same kVA rating it will be smaller, cheaper and lighter than three individual single phase transformers connected together because the copper and iron core are used more effectively. The methods of connecting the primary and secondary windings are the same, whether using just one Three Phase Transformer or three separate single phase transformers.

The primary and secondary windings of a transformer can be connected in different configuration as shown to meet practically any requirement. In the case of three phase transformer windings, two forms of connection are possible: “star” (wye), “delta” (mesh). The combinations of the three windings may be with the primary delta-connected and the secondary star-connected, or star-delta, star-star or delta-delta, depending on the transformers use.

Circuit Diagram:



Procedure:

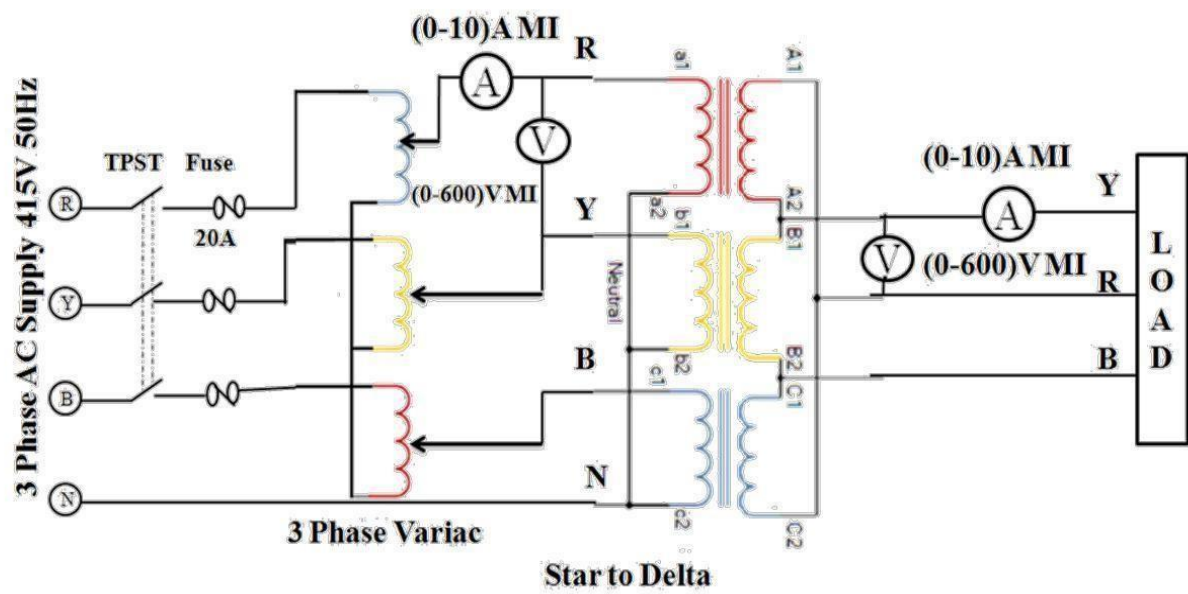
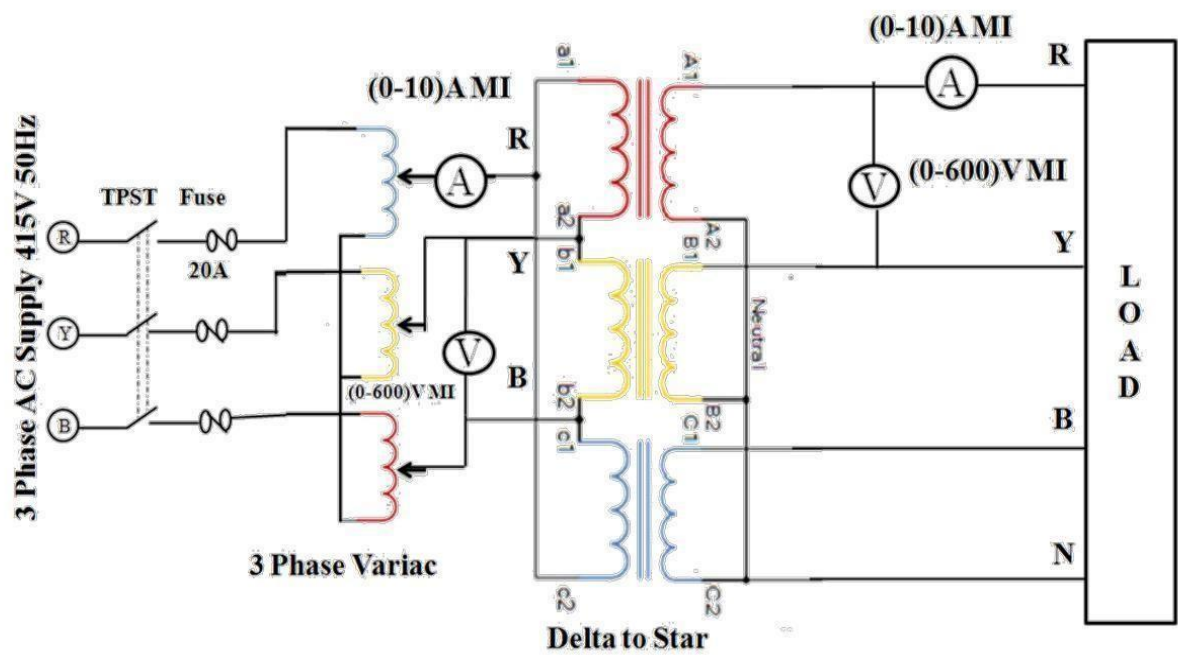
1. All the connections are done as per the circuit diagram for star-star connected transformer.
2. Before closing TPST switch, check the 3 phase Variac in minimum position.
3. By using 3 phase Variac apply rated voltage to the circuit.
4. Vary the R-Load to an appropriate value (10A) in steps, and note down the primary and secondary current and voltage for each step.
5. Reduce the voltage of the 3 phase Variac power supply to 0V and switch-off the supply.
6. Again connect the circuit for Delta-Delta connected transformer and follow the 2,3,4 and 5 steps.

Observation table:**Delta-Delta**

S. No	Primary Voltage V (in Volts)	Primary Current I (in amps)	Secondary Voltage V (in Volts)	Secondary Current I (in amps)
1				
2				
3				
4				

Star – Star

S. No	Primary Voltage V (in Volts)	Primary Current I (in amps)	Secondary Voltage V (in Volts)	Secondary Current I (in amps)
1				
2				
3				
4				



Procedure:

1. All the connections are done as per the circuit diagram for Delta-star connected transformer.
2. Before closing TPST switch, check the 3 phase variac in minimum position.
3. By using 3 phase variac apply rated voltage to the circuit.
4. Vary the R-Load to an appropriate value (10A) in steps, and note down the primary and secondary current and voltage for each step.
5. Reduce the voltage of the 3 phase variac power supply to 0V and switch-off the supply.
6. Again connect the circuit for star -Delta connected transformer and follow the 2,3,4 and 5 steps.

Observation table: Delta- Star

S. No	Primary Voltage V (in Volts)	Primary Current I (in amps)	Secondary Voltage V (in Volts)	Secondary Current I (in amps)
1				
2				
3				
4				

Star -Delta

S. No	Primary Voltage V (in Volts)	Primary Current I (in amps)	Secondary Voltage V (in Volts)	Secondary Current I (in amps)
1				
2				
3				
4				

Result:**Discussion Questions:**

1. What is working principle of a transformer?
2. What are the different losses in a transformer?
3. What are the different types of transformers

Outcome:

1. Students will be able to know about star-Delta, Delta-Star, Star-Star, Delta-Delta Connections in a Transformer.
2. Students will be able verify the relation between voltages and currents in star-Delta, Delta- Star, Star-Star, Delta-Delta Connections in a Transformer.

Experiment No:_____

Date:_____

MEASUREMENT OF **ACTIVE AND REACTIVE POWER** IN A BALANCED THREE PHASE CIRCUIT

Objective:

To measure the Active power (True Power [or] Real Power) and Reactive power for a balanced load in three phase circuit.

Apparatus:

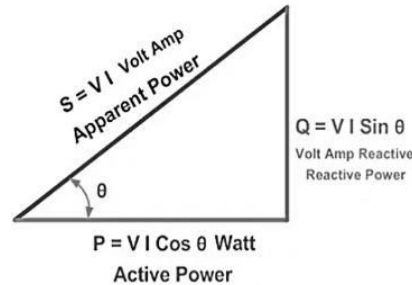
S.No	Name of the Equipment	Range	Type	Quantity
1	Voltmeter	(0-300)V	MI	1
2	Ammeter	(0-2)A	MI	1
3	Wattmeter	150/300/600V, 10/20A	UPF	1
		75/150/300V, 2.5/5A	LPF	1
4	Rheostat	100 Ω / 2A	wire wound	3
5	Inductive Load	25/50mH, 2A	----	3
6	3 Phase Variac	415V/(0-470)V, 4A	-	1
7	Connecting Wires	--	-	As per need
8	SPDT Switch	-	-	-

Theory:

Whenever the electric current flows it does some work. This work generates power that can be stored and used as energy. The rate at which electrical energy is consumed or dissipated into other forms of energy is called electrical power. When an alternating supply is given to a circuit, the expression of power i.e, real power is given as $P = V I \cos\Phi$. Where $\cos\Phi$ is the power factor.

Active power is the actual amount of power being used, or dissipated, in a circuit. It is denoted by P, and it is measured in watts(W). Reactive power is the power that continuously bounce back and forth between source and load. It is useless power denoted by Q and measures in volt ampere reactive (var). Apparent power is the combination of reactive power and true power is called apparent power, and it is the product of a circuit's voltage and current, without reference to phase angle. It is represented by S and units are Volt Ampere (VA).

Power Triangle is represented as below:



Generation of three phase power is more economical than generation of single-phase power. In three phase electric power system, three phase voltage are equal in magnitude and displaced by 120 degrees. Below are the expressions for three phase power:

$$\text{Active power (P)} = \sqrt{3} V I \cos \Phi$$

$$\text{Reactive power (Q)} = \sqrt{3} V I \sin \Phi$$

To measure active and reactive power there are different methods. One of the method is single wattmeter method .This method is used only when the load is balanced. To measure active power circuit is connected as per circuit diagram 8.1 and power is measured using Wattmeter (W) which is connected between two different phase and by using SPDT reading are as below

W_1 : Wattmeter reading when SPDT is at 'Y' (phase) Position [$W_1 = \sqrt{3} V_L I_L \cos(30 - \Phi)$]

W_2 : Wattmeter reading when SPDT is at 'B' (phase) Position [$W_2 = \sqrt{3} V_L I_L \cos(30 + \Phi)$]

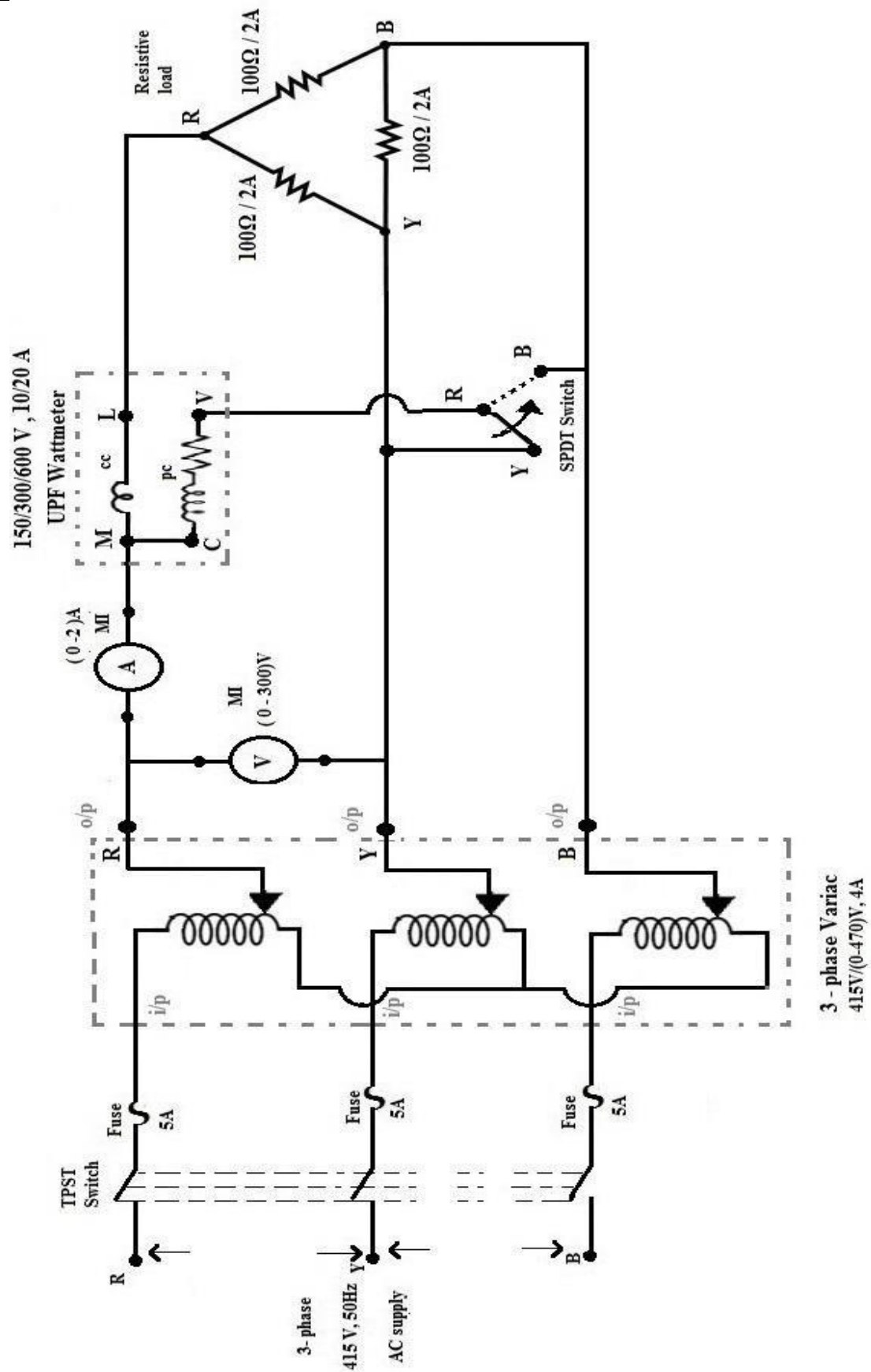
Total power is given by $P = W_1 + W_2$ and phase angle is $\Phi = -1 \left\{ \sqrt{3} \left[\frac{(1-2)}{(1+2)} \right] \right\}$

To measure reactive power circuit is connected as per circuit diagram 8.2 where the current coil is connected to one phase and pressure coil is connected across other two phases.

Reading of wattmeter = $V_{YB} I_R \cos (90 + \Phi) = \sqrt{3} VI \cos(90 + \Phi) = -\sqrt{3} VI \sin \Phi$

And phase angle is $\Phi = -1 \left(- \right)$

Circuit Diagram 8.1: For measurement of Active Power



Procedure: (For measurement of active power)

1. Make the connections as per the circuit diagram 8.1.
2. Make sure three phase Variac is at minimum position and note down multiplication factor of meters
3. Switch on the supply using TPST.
4. By adjusting 3phase Variac apply rated current (2A) in Ammeter.
5. Note voltmeter, ammeter & wattmeter (W1) reading when the SPDT is at position 'Y'.
6. Now move SPDT to position 'B' and note down wattmeter (W₂) reading.
7. Set Variac to minimum position and switch off TPST.
8. Calculate total active power

Observation Table: For measurement of Active power(P)

S.No	Line Voltage V _L (V)	Line Current I _L (A)	W ₁	W ₂	P = W ₁ +W ₂	$\Phi = \tan^{-1} \left\{ \sqrt{3} \left[\frac{W_2 - W_1}{W_1 + W_2} \right] \right\}$	Power factor = cos (Φ)

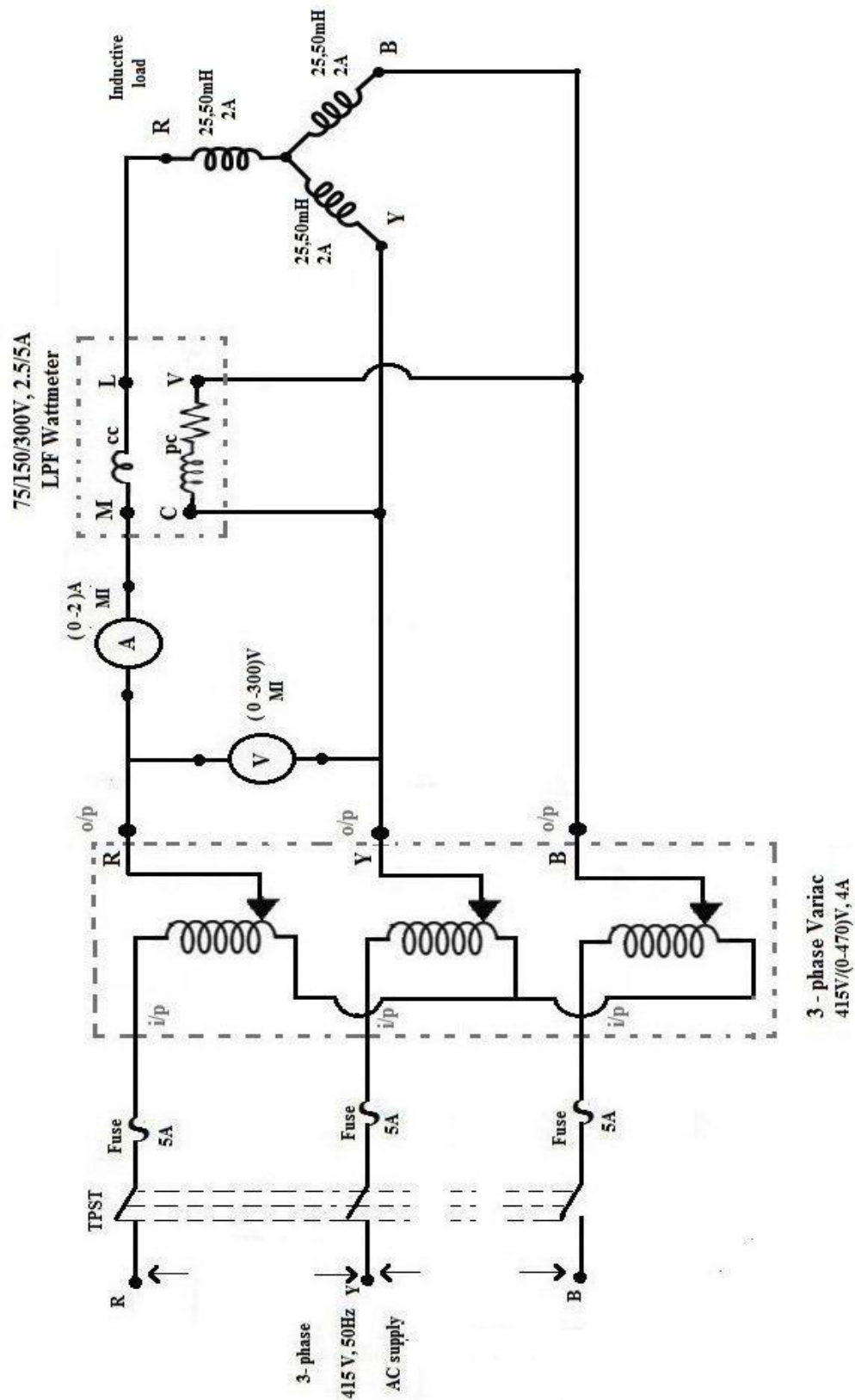
Note:

W₁ : Wattmeter reading when SPDT is at 'Y' (phase) Position

W₂ : Wattmeter reading when SPDT is at 'B' (phase) Position

$$\Phi = \tan^{-1} \left\{ \sqrt{3} \left[\frac{W_2 - W_1}{W_1 + W_2} \right] \right\}$$

Circuit Diagram 8.2: For measurement of Reactive Power



Procedure: (For measurement of Reactive power)

1. Make the connections as per the circuit diagram 8.2.
2. Make sure three phase Variac is at minimum position and note down multiplication factor of meters
3. Switch on the supply using TPST.
4. By adjusting 3-phase Variac apply rated current (2A) in Ammeter.
5. Note voltmeter, ammeter & wattmeter reading (W).
6. Adjust Variac to minimum position and switch off TPST.
7. Calculate total Reactive power

Observation Table: For Measurement of Reactive power (Q)

S. N o	Line Voltage V_L (V)	Line Current I_L (A)	Watt meter (P) (watts)	Power Factor $\cos \phi = \frac{\quad}{\sqrt{3}}$	Phase Angle = $\phi = \cos^{-1} \left(\frac{\quad}{\sqrt{3}} \right)$	Reactive Power Q = $\sqrt{3} \sin \phi$

Result:**Discussion Questions:**

1. Define power and what are the units of voltage, current and power.
2. What is complex power, active power and reactive power and write their units?
3. Write the active power equation for star network.
4. What are balanced and unbalanced loads?
5. Write the full form of SPDT, DPST, and TPST?

Outcome:

1. Students will be able to measure the practical values of active and reactive power for balanced load in three phase circuit.
2. Students will be able to understand the relation between Active Power and Reactive Power.

BASIC ELECTRICAL ENGINEERING LAB

For laboratory course(s), there shall be a **Continuous Internal Evaluation (CIE)** during the semester for **40 marks**, and **Semester End Examination (SEE) for 60 marks**.

A detailed break up of 40 marks for CIE is given below:

1. **A write-up on day-to-day** experiment in the laboratory (in terms of aim, components/procedure, expected outcome) which shall be evaluated for **10 marks**.
2. 10 marks for viva-voce (or) tutorial (or) case study (or) application (or) poster presentation of the course concerned.
3. **Internal practical examination** conducted by the laboratory teacher concerned shall be evaluated for **10 marks**.
4. **The remaining 10 marks are for Laboratory Project**, which consists of the Design (Or) Software / Hardware Model Presentation (or) App Development (or) Prototype Presentation submission which shall be evaluated after completion of laboratory.

In the Semester End Examination held for 3 hours, total 60 marks are divided and allocated as

shown below:

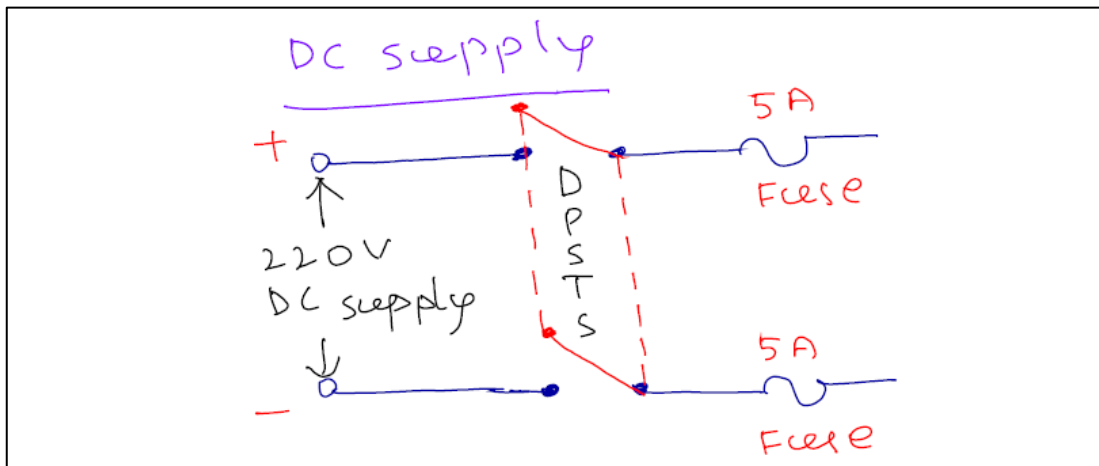
1. 10 marks for write-up
2. 15 for experiment/program
3. 15 for evaluation of results
4. 10 marks for presentation on another experiment/program in the same laboratory course and
5. 10 marks for viva-voce on concerned laboratory course

TOTAL MARKS = CIE + SEE = 14 + 21

PASS MARKS = 40

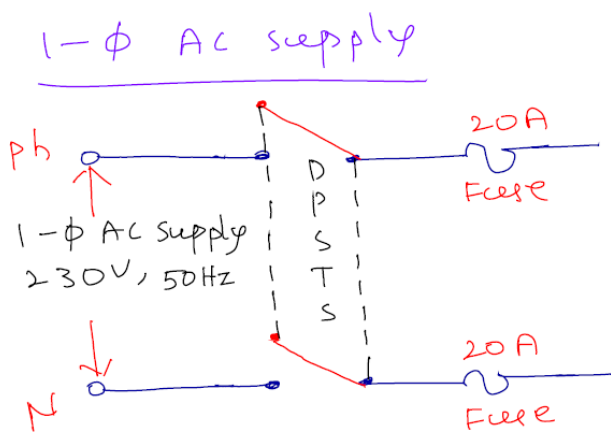
TYPES OF SUPPLY

1. DC SUPPLY:

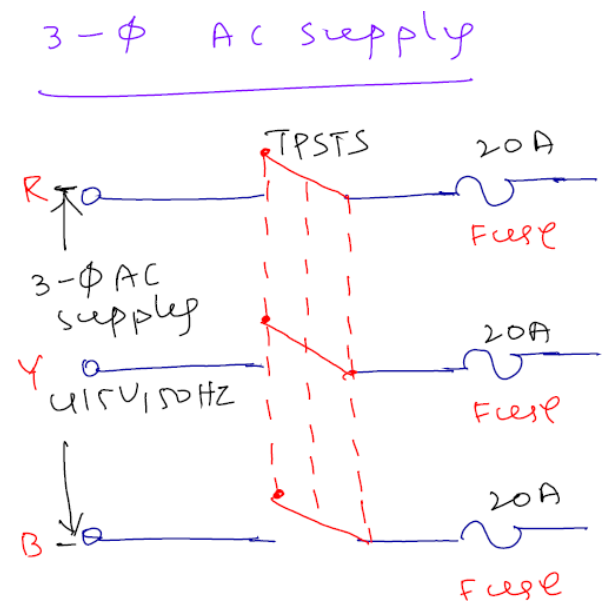


2. AC SUPPLY:

SINGLE-PHASE: 1- ϕ

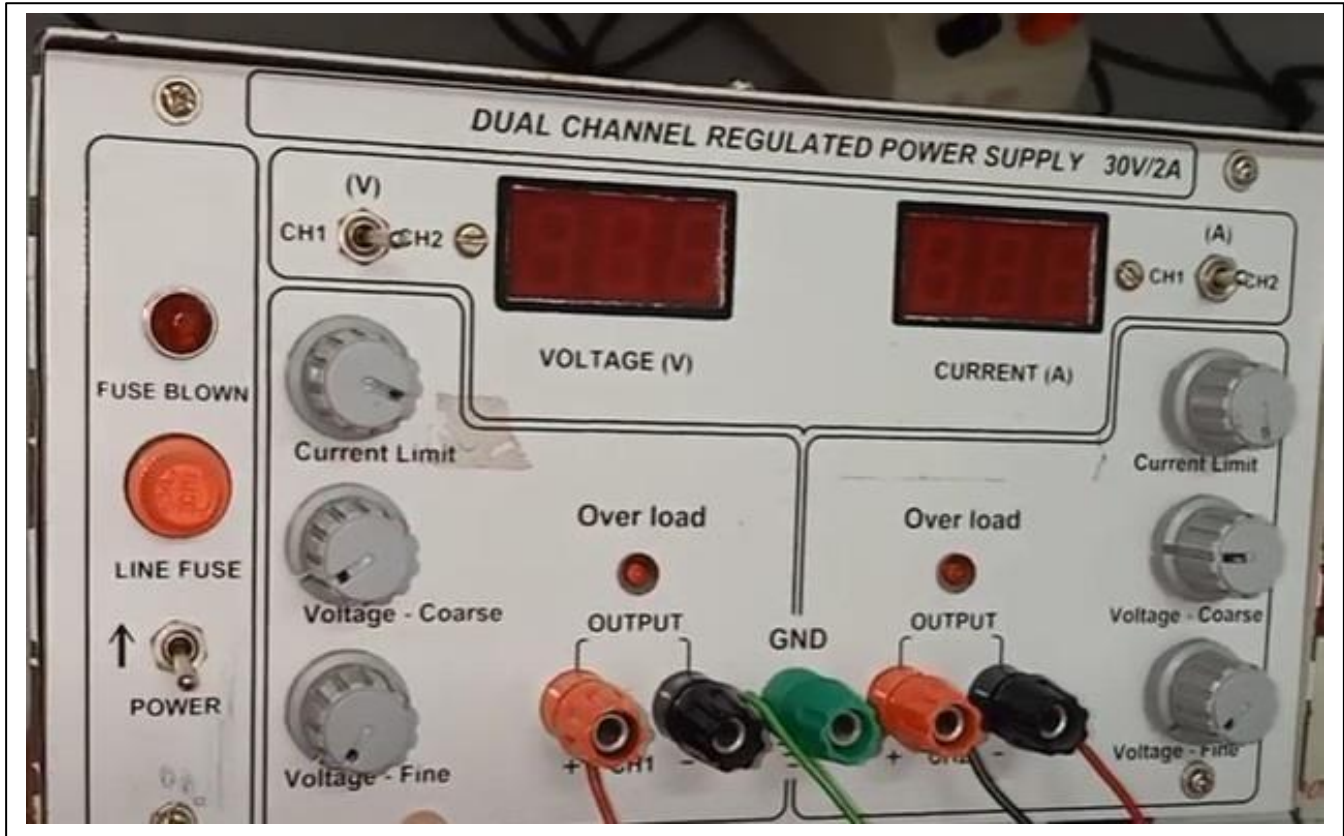


THREE-PHASE: 3- ϕ

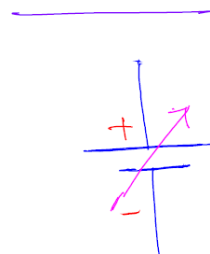


➤ APPARATUS USED IN LABORATORY

❖ Regulated Power supply:



DC RPS



DC RPS
(0-30)V, 2A

- ✓ We have several electrical and electronic circuits that require a DC power supply to operate.
- ✓ But, due to some technical and economical limitations, we cannot use conventional sources of DC power such as batteries and DC generators.
- ✓ In this situation, **we use an electronic circuit that can convert AC power into DC power, such a circuit is known as rectifier.**

Regulated Power Supply

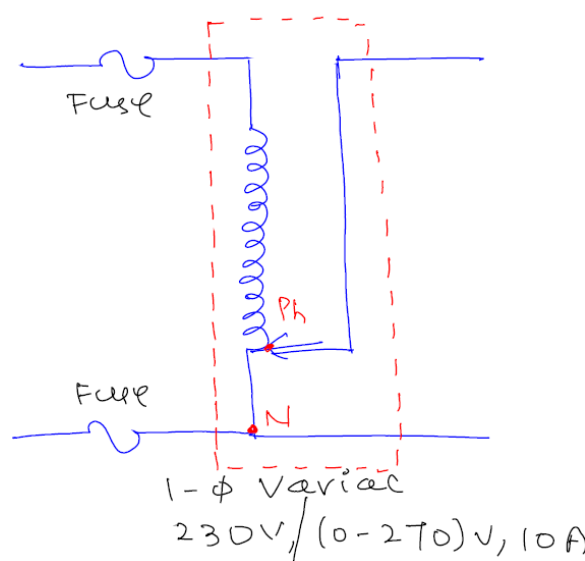
- ✓ An electronic circuit that **produces a stable DC voltage** of fixed value across the load terminals irrespective of changes in the load is known as regulated power supply.
- ✓ Thus, the primary function of a regulated power supply is to **convert an AC power into a steady DC power.** The regulated power supply is sometimes also called as a linear power supply.

SINGLE-PHASE VARIAC (OR) AUTO TRANSFORMER

- ✓ A variable autotransformer is a single-coil transformer in which two portions of the **same coil are used as the primary and the secondary**.
- ✓ A variac is a type of autotransformer that provides **adjustable AC output voltage**.
- ✓ Single phase variac transformers are widely used in different industries to **regulate voltage and current**.

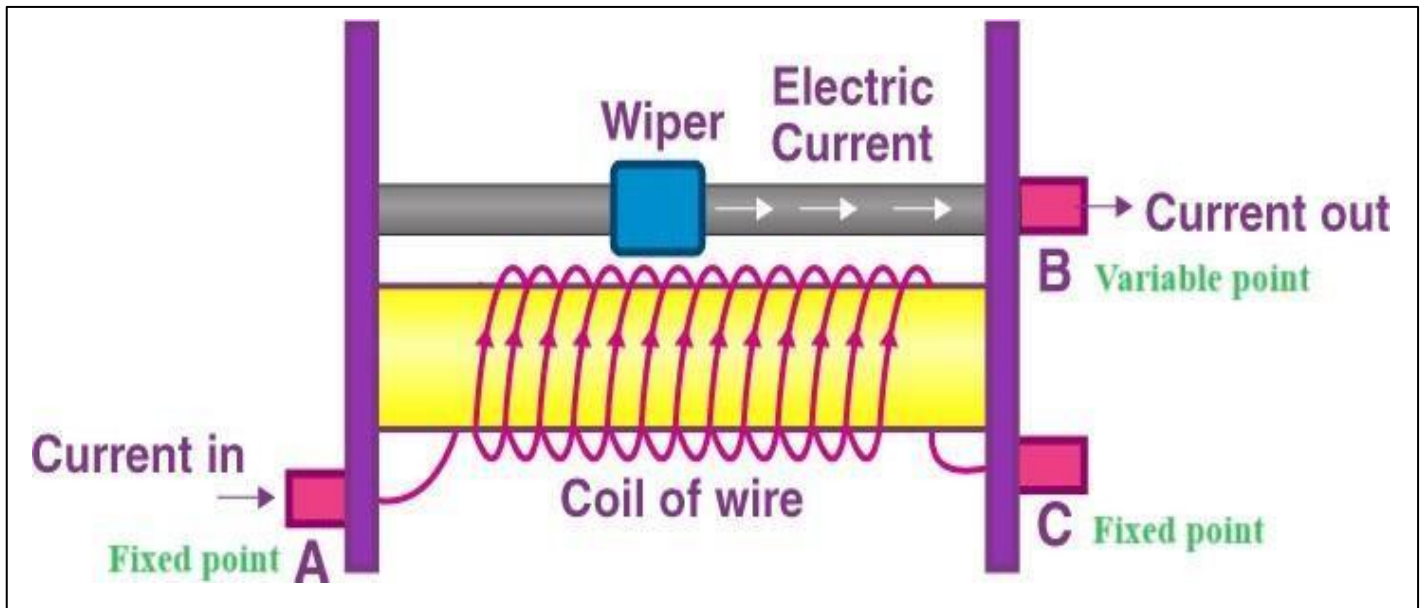


1- ϕ variac

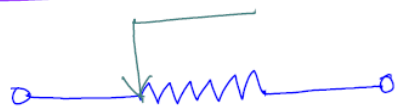


Rheostat

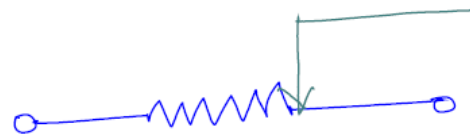
- ✓ **A rheostat is a variable resistor which is used to control current.**
- ✓ They are able to vary the resistance in a circuit without interruption.
- ✓ A variable resistor which is used for controlling the flow of electric current either by increasing or decreasing the resistance.
- ✓ **It finds applications in motor speed control, light control,** etc as it can handle higher current and voltage.
- ✓ It is not necessary to use all the three terminals.



Rheostat:



Minimum
resistance position



Maximum
resistance position

FUNCTION GENERATOR:

- ✓ Function Generator is basically a signal generator that produces different types of waveforms at the output.
- ✓ It has the ability to produce waveforms such as sine wave, square wave, a triangular wave, sawtooth wave etc. An adjustable frequency range is provided by the function generator which is in the range of some Hz to several 100KHz.
- ✓ Initially DC OFFSET should be off and Amplitude should be maximum.



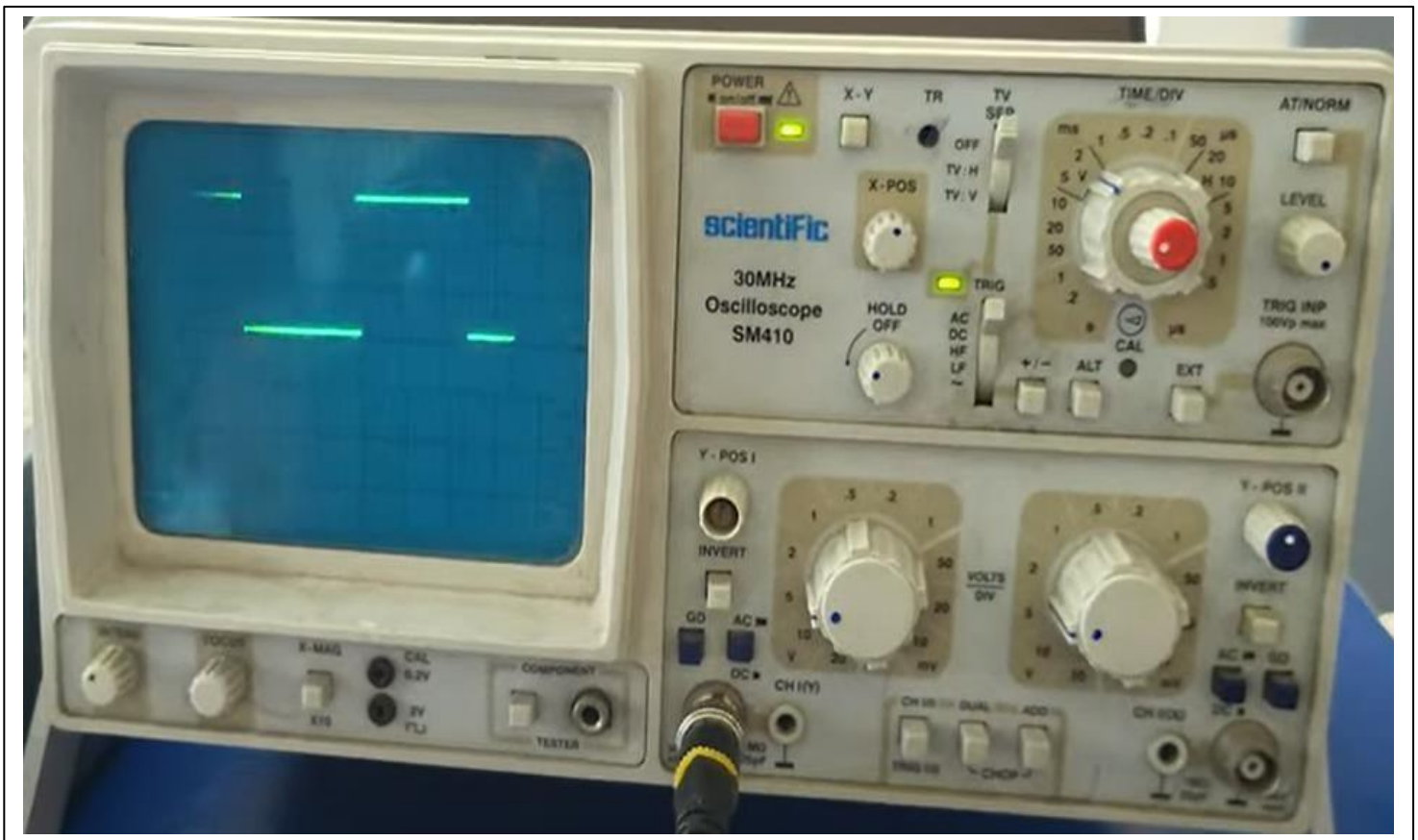
Function Generator

(0-1)
MHz



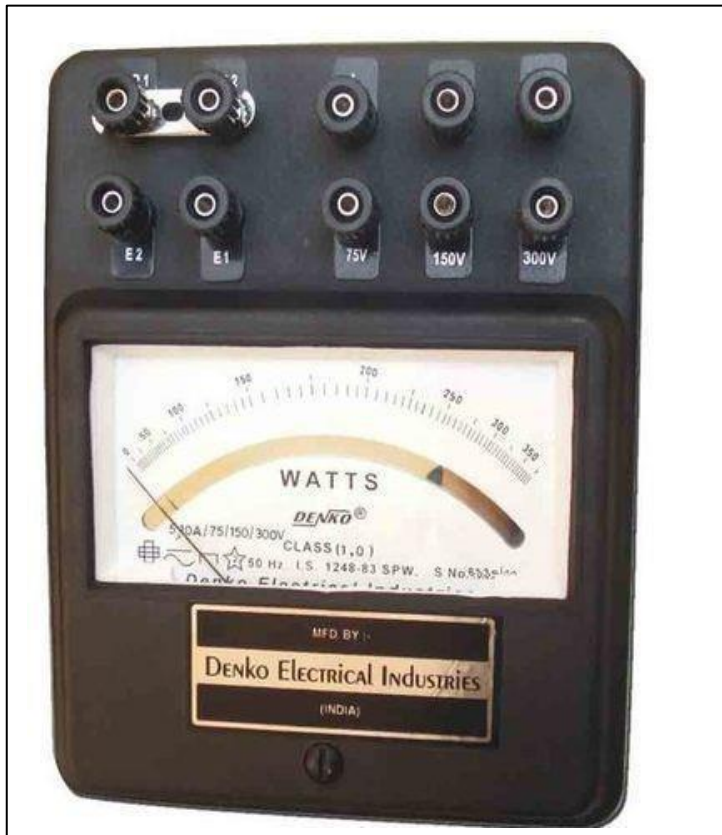
Cathode ray oscilloscope (CRO)

The cathode ray oscilloscope (CRO) is a type of electrical instrument which is used for showing the measurement and analysis of waveforms and others electronic and electrical phenomenon. It is a very fast X-Y plotter shows the input signal versus another signal or versus time. The CROs are used to analyse the waveforms, transient, phenomena, and other time-varying quantities from a very low-frequency range to the radio frequencies.

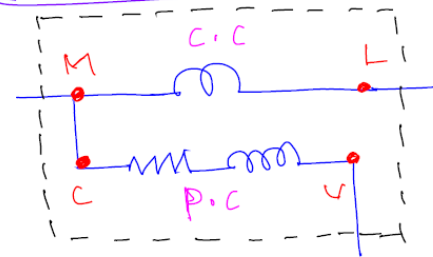


Wattmeter

The wattmeter is an instrument for measuring the electric active power in watts of any given circuit.



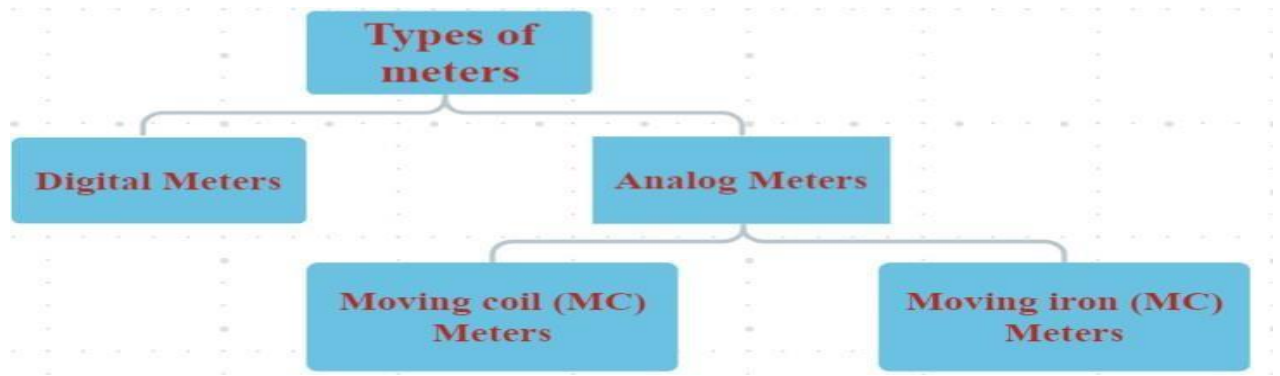
Wattmeter



75/150/300V, 5/10A, 0.8 LPF

- ✓ It consists of two coils, one is used to measure current, and another one is used to measure voltage and four terminals.
- ✓ The coil is used to measure current is called as Current coil (CC).
- ✓ The coil is used to measure voltage is called as Voltage coil (VC) or Pressure coil (PC).
- ✓ Four terminals are M, L, C, & V.
M = Main, L = Line, C = common & V = Voltage.

MEASURING INSTRUMENTS



DIGITAL METER:

- ✓ A digital meter is a meter that gives a separate reading, in the form of a decimal number, for each given input quantity.
- ✓ A digital meter is one that displays values as numerical values rather than as the position of a meter on a relative scale.
- ✓ In our lab digital meters are used for Cycle-I experiments.

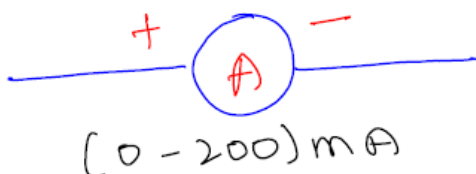
Digital Ammeter:

A digital ammeter is an instrument that measures the flow of an electrical current in a circuit in the form of a decimal number. The measurements are given in amperes.

- ✓ Ammeter always connected in series with the circuit element.



Digital Ammeter



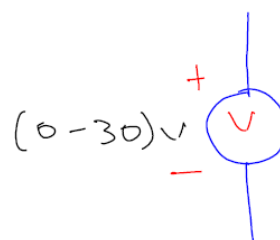
Digital Voltmeter:

A digital voltmeter is an instrument that measures the voltage across the element in a circuit in the form of a decimal number. The measurements are given in volts.

- ✓ Voltmeter always connected in parallel with the circuit element.



Digital voltmeter



ANALOG METER:

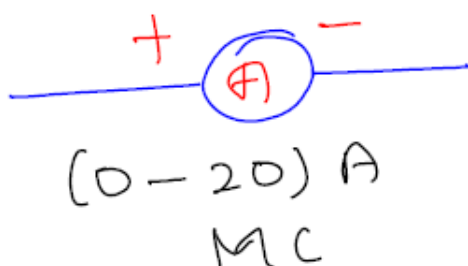
- ✓ These meters are used to measure quantities like voltage, current, energy, and power.
- ✓ An analog meter has a pointer or dial that shows the magnitude of the quantity being measured.

i. MOVING COIL TYPE METER:

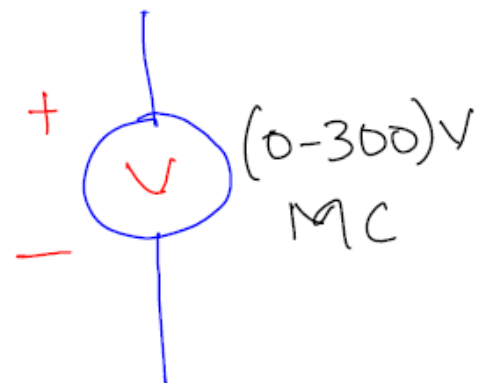
- ✓ Moving coil instruments can be used for DC supply only.
- ✓ The moving coil instrument has a uniform reading scale.

i. MC Ammeter

MC Ammeter

**ii. MC Voltmeter**

MC voltmeter



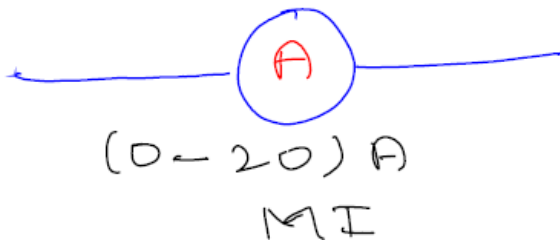
ii. MOVING IRON TYPE METER:

- ✓ Moving iron instruments can be used for both DC supply and AC supply.
- ✓ But most used for AC supply.
- ✓ The moving coil instrument has a non-uniform reading scale.

ii. MI Ammeter



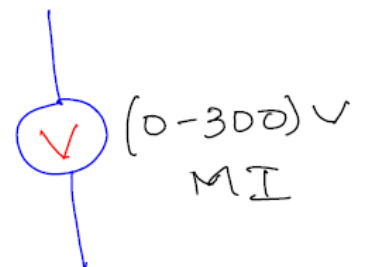
MI Ammeter



iii. MI Voltmeter



MI voltmeter



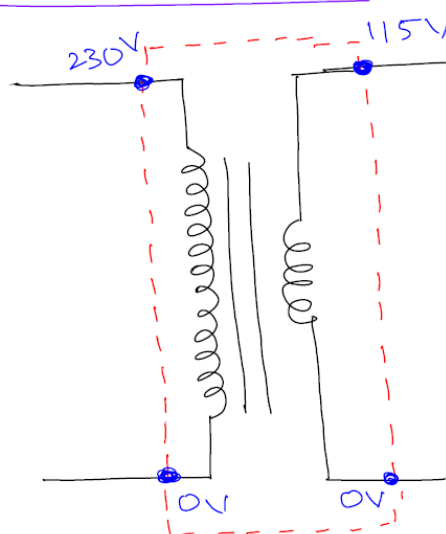
TRANSFORMER:

It is a static device which is used to transfer electrical energy or power from one circuit to another without change in frequency.

- ✓ It has two windings, one primary and another one secondary.
- ✓ Primary winding is connected to AC supply.
- ✓ Whereas secondary winding is connected to load.



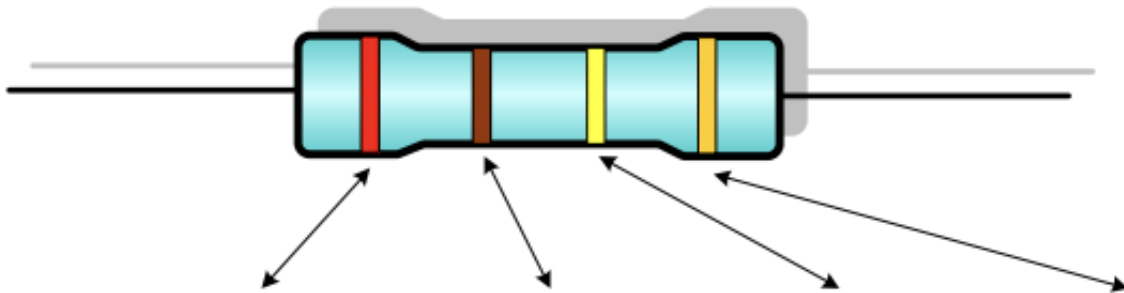
1- ϕ Transformer



1- ϕ Transformer

230V | 115V

2KVA

RESISTOR COLOR CODE:**4-band Resistor**

Color	1 st band value	2 nd band value	Multiplier	Tolerances
Black	0	0	$\times 1$	
Brown	1	1	$\times 10$	$\pm 1\%$
Red	2	2	$\times 100$	$\pm 2\%$
Orange	3	3	$\times 1000$	$\pm 3\%$
Yellow	4	4	$\times 10,000$	$\pm 4\%$
Green	5	5	$\times 100,000$	$\pm 0.5\%$
Blue	6	6	$\times 1,000,000$	$\pm 0.25\%$
Violet	7	7	$\times 10,000,000$	$\pm 0.10\%$
Grey	8	8	$\times 100,000,000$	$\pm 0.05\%$
White	9	9	$\times 1,000,000,000$	
Gold			$\times 0.1$	$\pm 5\%$
Silver			$\times 0.01$	$\pm 10\%$
No band				$\pm 20\%$

Formula to calculate resistance value:

$ABC \times 10^D = \text{for Five band resistors}$

$AB \times 10^C = \text{for four band resistors}$

